Performance Pay for Health Insurers: Evidence from Colombia's High-Cost Account*

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November 3, 2026

Abstract

Designing payments for health insurers is central to the proper functioning of health systems. This paper studies the impacts of performance pay for health insurers leveraging its introduction in Colombia for HIV/AIDS in 2016 and cancer in 2019. Using individual-level enrollment and health claims, we find that performance pay improved disease management for both conditions, increasing healthcare spending per patient by 25% and 28%, respectively. Heterogeneity analyses show that cost and payment uncertainty mediate these impacts: treatment effects are more than three times as large for health conditions with lower cost variance, and twice as large among insurers facing lower variance in the likelihood of getting paid.

Keywords: Performance Pay, Health insurance, Moral hazard, Risk selection.

JEL codes: I11, I13, I18, L13.

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1 Introduction

Designing payments for health insurers is central to the proper functioning of health systems (Enthoven, 1993). These choices are high stakes for policymakers because they shape whether insurers select against costly patients, invest in better care, or participate in the market at all. Insurers' decisions in turn are consequential for patient health and mortality (Abaluck et al., 2021; Buitrago et al., 2025), suggesting that the impact of insurer payments extend beyond the fiscal sustainability of the insurance sector.

Most managed care systems rely on risk adjustment to set insurer payments, which targets selection but does not directly reward the quality of care patients actually receive. An alternative is performance pay, which ties compensation to measurable outcomes (Holmström, 1979), and has been widely adopted in other settings, from education to labor and government (Artz and Heywood, 2025; Chen, 2024; Cohodes et al., 2023; Khan et al., 2016; Barlevy and Neal, 2012; Goodman and Turner, 2010; Atkinson et al., 2009; Griffith and Neely, 2009; Jensen and Murphy, 1990).

Within healthcare, however, evidence on whether performance pay improves quality of care is mixed and comes mainly from providers (Burke et al., 2025; Gupta, 2021; Alexander, 2020; De Walque et al., 2015; Miller and Barbiarz, 2013). For insurers the evidence is scarce and largely discouraging (Fioretti and Wang, 2023; Markovitz et al., 2021; Layton and Ryan, 2015). This paper provides novel evidence on health insurer performance pay and characterizes the conditions under which these contracts can improve health outcomes and when they fail.

To derive intuition, we develop a simple conceptual framework of a risk-averse insurer that receives a capitated transfer from the government and a performance payment tied to a noisy quality measure, in the spirit of Holmstrom and Milgrom (1991). The model delivers three results that guide our empirical analysis. First, performance pay raises insurer effort, but uncertainty about whether that effort will be rewarded lowers it. Second, performance pay creates incentives to deter enrollment not only from consumers with negative expected profit but also from those for whom meeting the quality thresholds is most uncertain. Third, when the quality score aggregates across many health conditions, effort directed

at any one condition moves the score only in proportion to that condition's weight, so an aggregated contract induces less effort than one targeted at specific conditions. Together these results offer one explanation for why Medicare Advantage (MA) star ratings have failed to improve quality of care. Many MA metrics, such as reducing the risk of falling or maintaining physical health, are not tied to any particular condition, which both dilutes incentives and makes the mapping from effort to payment hard to predict.

We provide causal evidence of our model's predictions exploiting the introduction of performance pay for health insurers in Colombia. Under this mechanism, known as the High-Cost Account (HCA), insurers receive compensation for enrollees with targeted conditions only if they meet specified quality-of-care standards. The HCA was introduced for HIV/AIDS in 2016 and for several cancers (breast, cervical, colorectal, stomach, and lung cancer, leukemia, and non-Hodgkin lymphoma) in 2019.[1] Funds are redistributed from insurers with below-average prevalence of the targeted conditions to those with above-average prevalence that meet the quality thresholds (Decree 2699 of 2007), making the mechanism budget-neutral for the government.

Our data span 13 years of individual-level enrollment records and health claims from all individuals enrolled in Colombia's contributory health system, roughly 24 million people per year from 2011 to 2023.[2] We study the effects of the policy for HIV/AIDS and cancer separately given their distinct patterns of healthcare use. For both conditions we use a cohort event study design. The key observation motivating our cohort approach is that healthcare utilization is concentrated in the first three years following diagnosis. Comparing within-person trends in healthcare use would therefore confound the natural disease progression with the policy's effect. Instead, we compare cohorts of patients diagnosed with targeted diseases before and after the policy, and further compare them against cohorts diagnosed with untargeted diseases over the same period to net out

[1] Quality standards include thresholds for disease prevalence and incidence, viral load monitoring, and medication adherence, among others.

[2] The contributory system covers individuals who pay payroll taxes and their dependents. The low-income population unable to pay taxes are covered by the subsidized system, which is fully funded by the government. Insurance coverage in Colombia is near universal. We do not have health claims data for the subsidized system.

contemporaneous trends unrelated to the policy.[3]

We find that performance pay for cancer increased annual healthcare spending per patient by 27.5% relative to the baseline of 7.7 million pesos ($2,600). The spending increase was concentrated in outpatient care and consistent with improved disease management. Patients diagnosed with targeted cancers after the policy had 12.4% higher rates of chemotherapy, radiotherapy, and cancer-related surgeries relative to baseline, suggesting that, prior to the introduction of performance pay, some patients had gone without appropriate cancer treatment or faced significant delays in receiving it.

Heterogeneity analyses confirm the theoretical predictions. First, treatment effects on spending and care quality outcomes are more than three times as large for health conditions with below-median variance in healthcare costs, suggesting that risk-averse insurers concentrate effort on more homogeneous diseases. Second, using a measure of each insurer's pre-policy likelihood of receiving performance payments, we show that insurers with above-median likelihood exhibit treatment effects more than twice as large, consistent with effort increasing in the ease of meeting the quality thresholds. Third, treatment effects are twice as large among insurers with below-median variance in their predicted payment likelihood as their counterparts, in line with effort decreasing in revenue uncertainty.

Our findings are qualitatively similar for the HIV/AIDS policy. Performance pay increased annual healthcare spending among post-policy cohorts by 25.0% relative to the baseline of 4.1 million pesos ($1,390). As with cancer, this is consistent with improved disease management: the likelihood of claiming antiretroviral therapy (ART) medications rose by nearly 16.9%, and the likelihood of receiving vaccinations against bacterial and viral infections that are common complications of HIV/AIDS increased by 157.1%. Heterogeneity analyses also confirm that these treatment effects are driven by health conditions with below-median cost variance, consistent with our model's characterization of

[3]These untargeted cohorts are specially important to net out the effects of the COVID-19 pandemic which was declared by the World Health Organization in March 2020. If targeted diseases were not disproportionately affected by the pandemic compared to untargeted ones, then the comparison allows use to identify the true policy effect. Because the HIV/AIDS policy was introduced before the pandemic, its estimated effects offer additional evidence on the impact of performance pay, independent of pandemic-related confounds.

optimal effort and indicating that the results are not an artifact of any single disease.

Together, these findings raise the question of which channels insurers use to affect health outcomes for HIV/AIDS and cancer patients. Because insurers in Colombia compete mainly on their networks of covered providers (other elements of the plan are uniform across insurers), they have two levers over care: which facilities they authorize care at, and how quickly patients receive their medications, since dispensing takes place at insurer-owned facilities rather than at retail pharmacies. We examine both channels in our event study framework, using as outcomes the number of beds at the facility where patients receive care and an indicator for whether patients received any disease-specific medication in their year of diagnosis. We find that patients with targeted cancers were more likely to receive care at larger, more specialized facilities after the policy, with no corresponding effect for HIV/AIDS patients, since disease management in that case depends more on timely medication dispensing than on facility-based care. Indeed, the likelihood of receiving disease-specific treatment (ART medications, in the case of HIV/AIDS) within the year of diagnosis was nearly twice as large as the corresponding effect across the full three-year post-diagnosis window, indicating that wait times for medication dispensing shortened for these patients.

Our paper contributes to two literature. First, the literature on performance pay in healthcare (Miller and Barbiarz, 2013; Eijkenaar, 2012), which has largely focused on provider-level contracts and found mixed results (e.g., Gupta et al., 2023; Einav et al., 2022; Gupta, 2021; Ashraf et al., 2014; Mullen et al., 2010). For instance, Burke et al. (2025) find that performance pay for skilled nursing facilities in the U.S. failed to improve targeted quality metrics such as 30-day readmission rates, while Gupta (2021) and Gupta et al. (2023) show that hospital performance pay under Medicare reduced readmissions and improved care coordination with downstream providers. In developing countries, performance pay has been studied in the context of HIV (De Walque et al., 2015), finding increased testing following its introduction, and in the context of maternal and infant care (Basinga et al., 2011), finding higher rates of preventive care visits for children but no effect on prenatal care visits. We argue that one reason for these opposing findings is uncertainty in returns to effort: if the link between effort and measurable outcomes

is noisier in skilled nursing facilities than in hospitals, or in maternal care than in HIV management, then the resulting revenue uncertainty could dampen effort on rewarded tasks.

While most of the performance pay literature focuses on healthcare providers, a few exceptions study performance pay for insurers, including Meyers et al. (2025); Fioretti and Wang (2023); Markovitz et al. (2021); Layton and Ryan (2015). These authors show that star quality ratings in MA, which determine risk-adjusted payments for health plans, have not improved plan quality, but have strengthened incentives for risk selection and, in some cases, spurred plan entry. Vatter (2025) states that one reason star ratings fail to improve outcomes is that scores are too granular: coarser ratings are easier for consumers to interpret, which can promote competition between plans and raise quality. Our adds to this explanation the notion that performance pay can lower insurer effort when quality metrics are not tied to any particular health condition as many of the MA metrics that determine the start rating are.

2 Conceptual Framework

We present the full theoretical framework and all derivations in Appendix 1; here we give the main intuition. We consider a risk-averse health insurer that chooses effort along several dimensions of care quality to maximize profits under a performance pay contract. Effort represents actions that improve patient health outcomes, such as authorizing care at more specialized facilities or shortening the time to dispense medications. Premiums and all other plan characteristics are fixed exogenously. Let $\mathbf{e}_\theta = (e_{1\theta}, \ldots, e_{K\theta})$ denote effort directed at consumers of type θ, where $e_{k\theta} \geq 0$ is effort along quality dimension k. Insurer demand from type θ, $s_\theta(\mathbf{e}_\theta)$, is increasing and concave in each dimension of effort. Healthcare cost per enrollee, $C_\theta(\mathbf{e}_\theta)$, is random, with an expected value that is increasing and convex in each dimension of effort and a variance that scales with its mean, a well-documented feature of healthcare spending (Brown et al., 2014). Effort on one dimension does not affect demand, costs, or measured quality on any other dimension, which rules out the multitasking distortions emphasized by Holmstrom and Milgrom (1991).

The government evaluates the insurer on K disease-specific quality indicators and sets a threshold $q^*_{k\theta}$ for each. The insurer's true performance on indicator k is $\tau_{k\theta}(e_{k\theta})$, which is increasing in effort, but the government observes it with noise, $q_{k\theta} = \tau_{k\theta}(e_{k\theta}) + \varepsilon_{k\theta}$. The insurer is paid in proportion to the distance by which its measured score exceeds the threshold, aggregated across indicators with weights $w_k \geq 0$, $\sum_k w_k = 1$, and scaled by a per-capita fee p_θ:

$$B_\theta = p_\theta \sum_k w_k \max\{q_{k\theta} - q^*_{k\theta},\, 0\}.$$

Measurement noise $\varepsilon_{k\theta}$ makes the payment uncertain whenever the insurer's expected score is close enough to the threshold that noise can push it to either side.

The insurer evaluates uncertain profit by its mean minus a penalty proportional to its variance:

$$\Pi = \sum_\theta \left\{ \mathbb{E}[\pi_\theta] - \frac{\rho}{2} \operatorname{Var}(\pi_\theta) \right\}, \qquad \pi_\theta = \left(R_\theta - C_\theta + B_\theta\right) s_\theta(\mathbf{e}_\theta),$$

where R_θ is the risk-adjusted per-capita payment from the government and $\rho > 0$ captures risk aversion, which we interpret as the shadow cost of the solvency capital the insurer must hold against profit variance. Expanding this expression yields:

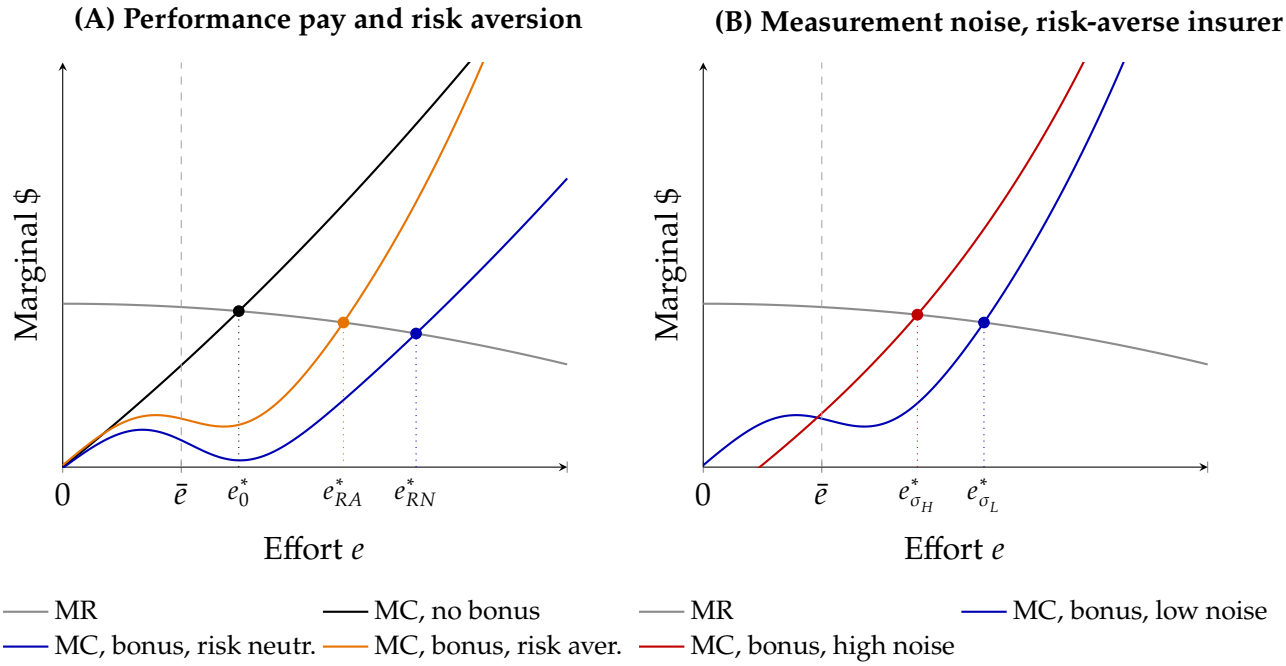
$$\Pi = \sum_\theta \left\{ \left[R_\theta - \mathbb{E}[C_\theta] + p_\theta \sum_k w_k m_{k\theta} \right] s_\theta - \frac{\rho}{2} s_\theta^2 \left[\operatorname{Var}(C_\theta) + p_\theta^2 \sum_k w_k^2 v_{k\theta} \right] \right\},$$

where $m_{k\theta} = \mathbb{E}\left[\max\{q_{k\theta} - q^*_{k\theta}, 0\} \right]$ and $v_{k\theta} = \operatorname{Var}\left[\max\{q_{k\theta} - q^*_{k\theta}, 0\} \right]$ are the mean and variance of the performance payment on indicator k.

From the insurer's objective function, effort raises expected revenue through two channels: first, it raises the probability of clearing each threshold, and hence the expected performance payment, and second it attracts enrollees. Effort also raises expected healthcare costs. For a risk-averse insurer, effort raises profit variance or the risk premium, for three reasons. First, higher expected costs mean more variable costs. Second, effort raises the variance of the performance payment, since below the threshold it converts a sure zero into an uncertain payment. Third, effort raises enrollment, which scales the insurer's

exposure to both sources of uncertainty. Performance pay therefore lowers the effective marginal cost of effort by the expected marginal bonus, while risk aversion raises it by the marginal risk premium.

FIGURE 1: Optimal Insurer Effort under Performance Pay and Risk Aversion



Note: Figure presents marginal revenue and effective marginal cost of effort on a single quality indicator under performance payments. The threshold effort $\bar{e}$ satisfies $\tau(\bar{e}) = q^*$. Panel A compares a risk-neutral insurer with a risk-averse one. The marginal revenue curve is in gray, the marginal cost curve without performance pay nor risk aversion in black, the marginal cost curve with performance pay but without risk aversion in blue, and the marginal cost curve with performance pay and risk aversion in orange. Panel B holds risk aversion fixed and compares low and high measurement noise. The marginal revenue curve is in gray, the marginal cost curve with low measurement noise in blue, and the marginal cost curve with high measurement noise in red.

Figure 1 illustrates the resulting comparative statics in the space of effort and marginal dollars. Panel A shows the marginal revenue curve in gray, the marginal cost curve without performance pay (denoted "bonus") nor risk aversion in black, the marginal cost curve with performance pay but without risk aversion in blue, and the marginal cost curve with performance pay and risk aversion in orange.

The first pattern to highlight is the shape of the effective marginal cost curves with performance pay. Performance pay reduces the effective marginal cost in a non-uniform way with respect to effort. The magnitude of the reduction is a function of the probability of clearing the thresholds, which follows the probability distribution of $\varepsilon_{k\theta}$. For low values of effort this probability is essentially zero, so the cost curves under performance

pay coincide with the curve without it. As effort approaches the threshold denoted as $\bar{e}$, the probability rises steeply, and the marginal bonus grows faster than the underlying marginal cost of care, so effective marginal cost falls. Once effort is well above $\bar{e}$, the probability equals one, so the curves grow at a constant rate.

Optimal effort is the effort level at which the insurer's effective marginal cost equals its marginal revenue. Under risk neutrality, performance pay raises optimal effort from e_0^* to e_{RN}^*. Risk aversion then reduces effort to e_{RA}^*, since the risk premium shifts effective marginal cost upward. The size of that shift, and hence the gap between the blue and orange curves, scales with ρ, so a sufficiently risk-averse insurer could end up exerting less effort under performance pay than without it.

Panel B shows the impact of the government's measurement noise for a risk-averse insurer. When the noise is low, the effective marginal cost has the same shape as in Panel A. When noise is high, the curve flattens because the probability of clearing the threshold rises slowly over a wide range of effort. The two curves cross at $\bar{e}$, where the probability of clearing the threshold equals one half regardless of noise. To the left of $\bar{e}$, higher noise lowers the effective marginal cost, since an insurer that expects to fall short can now clear the threshold by luck; to the right, higher noise raises this probability, since an insurer that expects to clear the threshold now risks missing it. Because optimal effort lies to the right of $\bar{e}$ in this panel, the second effect dominates and higher noise lowers optimal effort from $e_{\sigma_L}^*$ to $e_{\sigma_H}^*$.

The result in Panel B also informs the design of performance pay contracts. When quality scores are aggregated across several health conditions, effort on one condition will move the score only in proportion to that condition's weight in the score, so the insurer's effort has a smaller and less predictable effect on whether the threshold is cleared. A pooled scheme therefore dilutes the marginal return to effort and raises payment uncertainty.

The conceptual framework delivers several predictions that guide our empirical analysis: insurer effort under performance pay should be higher for conditions whose treatment costs are less variable, for insurers that are more likely to clear the quality thresholds, and for insurers whose payments are less uncertain.

3 Institutional Background

To test the theoretical predictions, we use data from Colombia's healthcare system. This system was established in 1993 and is organized around two schemes. The contributory scheme covers the population who pay payroll taxes and their dependents (roughly half of the country's population), while the subsidized scheme is fully government-funded and covers low-income populations (the other half of the population). Together they achieve near-universal coverage.

Within each scheme, private and public insurers offer a single standardized national health plan. Premiums are zero, cost-sharing rules are indexed to enrollees' monthly income, and the set of covered services is determined by the government uniformly across insurers. Competition therefore does not occur along the price or benefit dimensions but along the formation of provider networks. Individuals can enroll with any insurer operating in their municipality of residence.

Instead of premiums, insurers receive per capita risk-adjusted transfers from the government that control for the enrollees' sex, age, and municipality of residence. Because the formula is relatively coarse (Riascos et al., 2014), it fails to account for the higher costs associated with chronic conditions, making these enrollees predictably unprofitable (Serna, 2024).

Performance Pay Policy. To address the limitations of the coarse risk adjustment formula, the Colombian government introduced a complementary mechanism in 2007 known as the High-Cost Account (HCA). The HCA compensates insurers for the costs associated with specific high-burden diseases, initially covering end-stage renal disease without specifying quality metrics and expanding to HIV/AIDS in 2016 and to cancer in 2019 (breast, cervical, colon, stomach, prostate, leukemia, and non-Hodgkin lymphoma) while at the same time introducing care quality metrics for these conditions.[4]

The HCA operates as follows. All insurers contribute to a common pool of funds in proportion to their national market shares, regardless of whether they enroll patients

[4]See Resolution 1912 of 2015, Ministry of Health and Social Protection. Hemophilia has been covered under the HCA since 2016, though payments for this condition are not tied to quality metrics.

with targeted diseases. To determine the size of this pool of funds, the government first computes the prevalence of disease θ at insurer j and age group a as:

$$\rho_{\theta aj} = 100{,}000 \cdot \frac{D_{\theta aj}}{N_{aj}}$$

where $D_{\theta aj}$ is the number of patients with disease θ in age group a enrolled with insurer j, and N_{aj} is the total number of enrollees in age group a at insurer j. Population-level prevalence for age group a is then $\rho_{\theta a} = 100{,}000 \cdot \frac{\sum_j D_{\theta aj}}{\sum_j N_{aj}}$.

The differential prevalence of disease θ at insurer j and age group a is $f_{\theta aj} = (\rho_{\theta aj} - \rho_{\theta a}) \cdot N_{aj}/100{,}000$, and total differential prevalence at the insurer is $f_{\theta j} = \sum_a f_{\theta aj}$. Letting p_θ denote the per-capita payment for disease θ (computed as the average healthcare cost in the population), the common pool of funds is:

$$CF_\theta = \sum_j \mathbf{1}\{f_{\theta j} > 0\} \cdot f_{\theta j} \cdot p_\theta$$

and each insurer contributes $b_\theta = CF_\theta \cdot N_j / \sum_j N_j$, where $N_j = \sum_a N_{aj}$. Intuitively, the common pool of funds is the expected amount of money needed to equalize disease prevalence across insurers. By making *all* insurers contribute to the pool of funds, the HCA implicitly transfers resources from low- to high-prevalence insurers.

Funds are then redistributed exclusively to insurers that meet government-specified quality thresholds for each targeted disease. Following the notation from the conceptual framework in Section 2, Let $q^*_{\theta k}$ denote the threshold for the k-th quality indicator for disease θ, and let $q_{\theta kj}$ denote insurer j's measured value of that indicator. Population-weighted compliance with the k-th indicator at insurer j is defined as:

$$I_{\theta kj} = \mathbf{1}\{q_{\theta kj} > q^*_{\theta k}\}(q_{\theta kj} - q^*_{\theta k})N_j$$

and the share of compliance attributed to indicator k is $\bar{I}_{\theta kj} = I_{\theta kj} / \sum_j I_{\theta kj}$. Insurer j then

receives a payment equal to:

$$B_\theta = \sum_k w_{\theta k} \cdot \bar{I}_{\theta kj} \cdot CF_\theta \tag{1}$$

where $w_{\theta k}$ is the weight the government assigns to the k-th quality indicator. Hence, only insurers with high disease prevalence that meet the quality thresholds get paid by the HCA in proportion to how far they are from the thresholds, as in our theoretical framework.

For HIV/AIDS, the quality thresholds include targets for the share of pregnant women screened for HIV/AIDS, the share of people with HIV/AIDS on ART with an adequate recent viral load, and the share of patients with early detection of the disease, among others. For cancer, they include the share of women screened for breast cancer, the share of men screened for prostate cancer, and the share of cancer patients in remission, among others. For HIV/AIDS, contributions to the common pool were collected between November and December 2015, with funds redistributed to qualifying insurers in January 2016. For cancer, contributions were collected in December 2018 and redistributed in January 2019.[5]

4 Data

We use individual-level enrollment and health claims data for the roughly 24 million people covered by the contributory system between 2011 and 2023. The enrollment files are snapshots of enrollment in June of every year, reporting the individual's insurer, municipality of residence, sex, and age. The health claims data report all the healthcare services that an individual received every year (outpatient, inpatient, and prescriptions) and include information such as the insurer, provider, negotiated service price, procedure code, and diagnoses following the International Classification of Diseases (ICD) nomenclature.

The enrollment records have information from all insurers including those in the subsidized system. That is, we observe individuals who are ever enrolled in the contributory

[5] See Resolution 5036 of 2015 and Resolution 3339 of 2019.

system including for years in which they move into the subsidized system. In the health claims data, we only have information for insurers in the contributory system. So, if an individual moves to the subsidized system in a given year we do not observe their healthcare use that year.

A key challenge for our empirical design is defining the treatment group and treatment timing when individuals may be diagnosed with multiple diseases in different years. For example, an individual diagnosed with breast cancer in 2013 and lung cancer in 2019 would belong to the pre-policy cohort based on the first diagnosis but to the post-policy cohort based on the second. To avoid this ambiguity, we restrict the cancer treatment group to individuals diagnosed with exactly one targeted cancer type who have no HIV/AIDS diagnosis during the sample period.

For the HIV/AIDS analysis, the treatment group comprises individuals diagnosed with HIV/AIDS with no concurrent cancer diagnosis, but allowing these patients to potentially be diagnosed with any other chronic health condition. We exclude HIV/AIDS patients ever enrolled with insurers EPS013 and EPS003, which were terminated by the government around the time the HIV/AIDS policy was introduced; excluding them ensures that our outcomes are not confounded by the care disruptions associated with these terminations (Buitrago et al., 2025). Appendix 2 describes our data cleaning process and the procedure for classifying patients as diagnosed with a targeted condition using the claims data.

We construct the control group for the cancer policy from conditions that are clinically similar to the targeted cancers but excluded from the HCA. For instance, stomach cancer (ICD-10 code C16) is a targeted condition, so the control group includes other cancers of the digestive system, such as malignant neoplasms of the esophagus (C15), small intestine (C17), and liver and intrahepatic bile ducts (C22). Appendix Table 1 lists the ICD codes defining the treatment and control groups for cancer. To validate the construction of diagnoses from ICD codes, Appendix Figure 1 compares the prevalence of targeted cancers in our sample against statistics from the contributory system reported to the HCA, finding that the two are closely aligned.

Unlike the cancer setting, where the HCA generates within-cancer type variation and

thus a natural comparison group, the HIV/AIDS policy applies uniformly across all HIV/AIDS patients, leaving no untreated subgroup within the same disease category. Thus, to construct a control group we focus on diseases that share the chronic-care profile of HIV/AIDS (which requires continuous medication and viral load monitoring): tuberculosis, asthma, hypertension, diabetes, and epilepsy.[6] Nevertheless, even among these plausible comparators, there is no reason to expect each disease to track HIV/AIDS's pre-policy spending trend equally well. To address this, we use a synthetic control method to construct a weighted combination of these control diseases with weights chosen specifically to reproduce HIV/AIDS's pre-period total healthcare spending trajectory, age at diagnosis, and share of males.

We implement this matching procedure at the level of 3-digit ICD codes. For each 3-digit code, we compute the average healthcare spending across all patients in that diagnosis per year from 2011 to 2015, the average age, and the share of males. Weights on the control diseases are chosen by solving a constrained quadratic program that minimizes the distance between the characteristics of treated 3-digit codes and a weighted combination of the control diseases' characteristics, subject to the weights being non-negative and summing to one, and with a ridge penalty added to discourage the solution from concentrating weight on a single control disease (Abadie and L'hour, 2021).

5 Empirical Design

After defining treatment and control groups, we examine pre-policy outcome trends to motivate our empirical design. The top panel of Appendix Figure 2 shows that healthcare spending among both treated and control cancer patients is low in the years preceding diagnosis and trends upward gradually. At the time of diagnosis, spending spikes sharply before tapering off and returning to near pre-diagnosis levels by the third year post-diagnosis. Healthcare utilization for cancer patients is therefore heavily concentrated in the first three years following diagnosis. The bottom panel of Appendix Figure 2 reports corresponding spending trends for HIV/AIDS patients and their control conditions.

[6] Appendix Table 1 lists the ICD codes used to identify HIV/AIDS patients and its control conditions.

Consistent with their chronic nature, most diseases exhibit relatively flat post-diagnosis spending trajectories.

These patterns inform our empirical design. We use a cohort event study that compares cohorts of individuals diagnosed with a targeted disease before the policy against those diagnosed after, using at most three years of post-diagnosis data, where healthcare use is concentrated. A simple comparison of pre- and post-policy cohorts would be equivalent to an interrupted time series, which is vulnerable to confounding from contemporaneous changes unrelated to the HCA. Hence, we introduce the control group of untargeted diseases. Our estimator computes the difference in outcomes between targeted and untargeted cohorts diagnosed after the policy, subtracts the analogous difference from cohorts diagnosed before the policy, and conditions these comparisons on the calendar year relative to the policy. This approach yields a classic event study design, formally in the equation below:

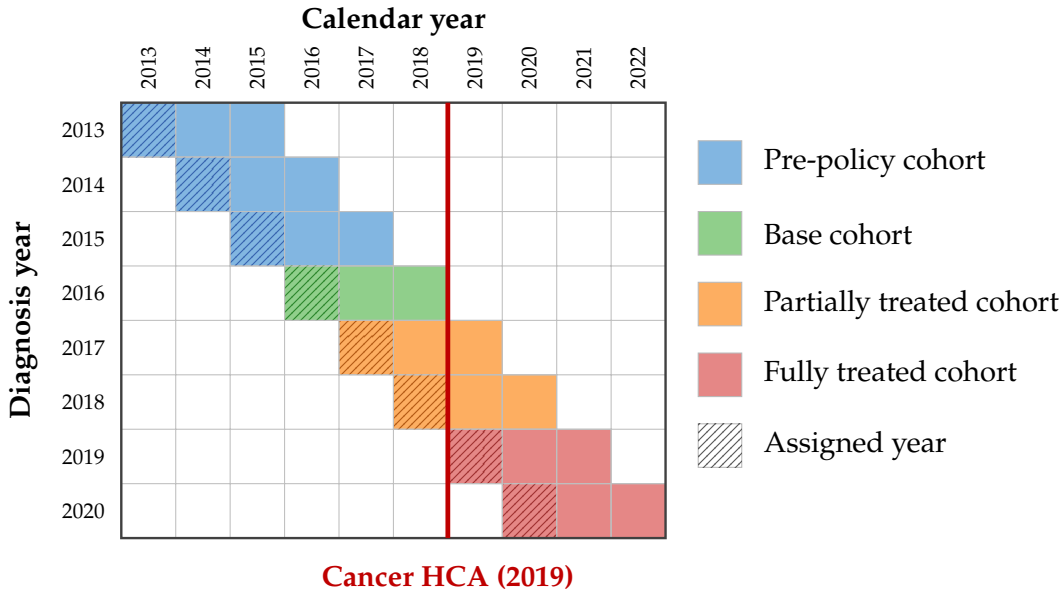
$$y_i = \sum_{k=-5, k\neq -1}^{4} \beta_k \mathbf{1}\{t - t^* = k\} T_{d(i)} + \gamma_{d(i)} + \delta_{t(i)} + \lambda_{j(i)} + \varepsilon_i \tag{2}$$

where i denotes an individual, $t(i)$ denotes their diagnosis year, t^* is the year of the HCA policy, $T_{d(i)}$ is an indicator for whether individual i has a diagnosis targeted by the HCA, $\gamma_{d(i)}$ is a diagnosis dummy (defined by the 3-digit ICD code), $\delta_{t(i)}$ is a cohort fixed effect, and $\lambda_{j(i)}$ is an insurer fixed effect. We cluster standard errors at the level of 3-digit ICD code, since this determines treatment by the policy, and estimate the regression separately for the cancer and HIV/AIDS policies.

Figure 2 illustrates the construction of our cancer sample for the cohort design. The design for the HIV/AIDS analysis is analogous. Each row corresponds to a diagnosis cohort and each column to a calendar year. For each patient, we compute the sum of outcomes over the three-year window following diagnosis and assign the result to the year of diagnosis. Cohorts whose outcome windows fall entirely before the policy (blue) serve as the pre-policy comparison group. The 2016 cohort (green) serves as the base cohort ($k = -1$), as it is the last whose outcome window falls entirely in the pre-period. The 2017 and 2018 cohorts are partially treated (orange), since portions of their outcome

windows fall within the post-policy period. Cohorts diagnosed in 2019 and later (red) are fully treated.

FIGURE 2: Cohort Event Study Design



Note: Figure presents the cohort design for our event study specification. Each row corresponds to a diagnosis cohort and column to a calendar year. Outcomes are averaged over the three-year window following diagnosis and assigned to the diagnosis year. The base cohort (2016) is the last cohort whose outcome window falls entirely in the pre-policy period. The 2017 and 2018 cohorts are partially treated. Cohorts diagnosed in 2019 and later are fully treated. The hatched cell in each row indicates the assigned (diagnosis) year.

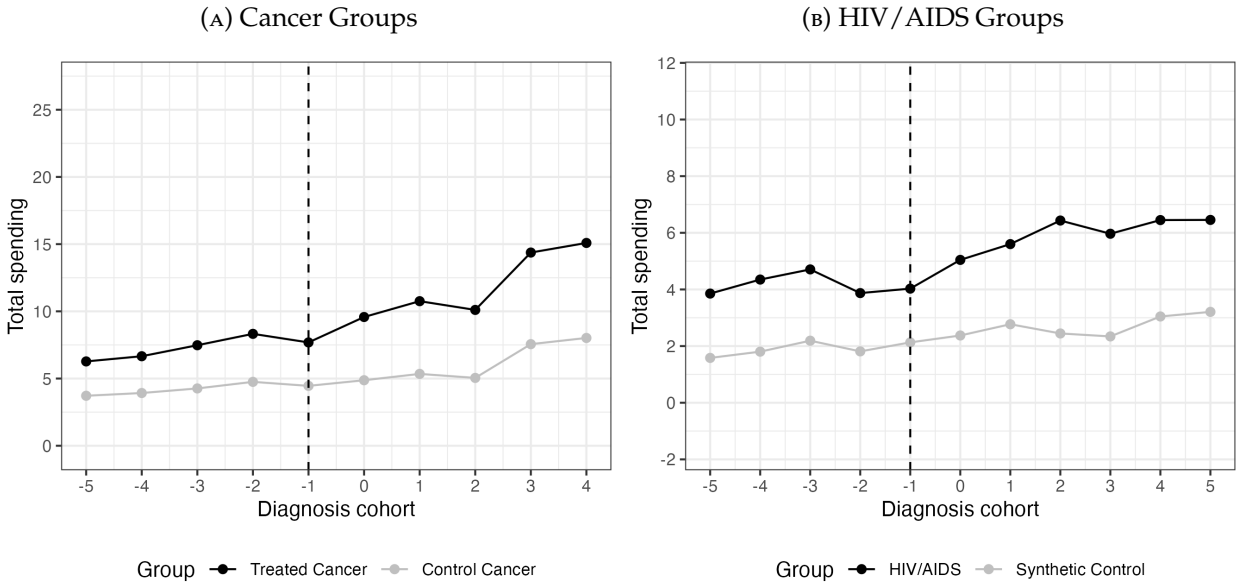
For HIV/AIDS, the pre-policy comparison group consists of cohorts diagnosed in 2011–2013. Cohorts diagnosed in 2014 and 2015 are partially treated, and 2016 marks the first fully treated cohort. For both analyses, we exclude patients diagnosed in 2022 and 2023 for whom outcomes are censored.

Appendix Table 3 presents summary statistics of treatment and control groups in the pre-policy period for the cancer analysis in Panel A and the HIV/AIDS analysis in Panel B. For cancer, the two groups are balanced in terms of the fraction of males and the pre-diagnosis healthcare spending. The treatment group gets diagnosed at slightly older ages and has higher total and disease-specific spending. HIV/AIDS patients are disproportionately male, have an average age of 36 years, and represent substantially higher healthcare spending than their comparator diseases.

Identification. Valid identification of the policy's effects rests on the following assumptions. The first assumption is the absence of contemporaneous policy confounders, such as changes in sample composition or the COVID-19 pandemic. Sample composition is particularly worrisome for a cohort analysis which naturally has different sets of patients in the pre- and post-policy periods. The pandemic confounds effects only if targeted diseases were disproportionately more likely to be affected by COVID-19. The second is assumption SUTVA. This assumption may be violated if insurers improve dimensions of care quality that apply broadly to all enrollees, in which case treatment effects would be underestimated. Conversely, if insurers divert effort away from untargeted conditions, control group outcomes would worsen and treatment effects would be overestimated. The third assumption is parallel pre-trends, requiring that outcomes for treated individuals would have evolved parallel to controls in absence of the HCA. We assess the plausibility of each of these assumptions below.

The first concern related to sample composition is that performance pay may alter diagnosis rates, changing the composition of post-policy cohorts. If the policy incentivizes earlier or more frequent screening, post-policy cohorts may systematically contain patients diagnosed at earlier disease stages relative to pre-policy cohorts, which would lead us to attribute differences to true improvements in care quality rather than differences in disease severity. We assess this potential source of bias by examining whether diagnosis rates and patient characteristics change around the policy introduction. Appendix Figure 5 compares age at diagnosis, number of diagnosed patients, share of male patients, and total pre-diagnosis healthcare spending between treatment and control groups for the cancer and HIV/AIDS policies. We find no relevant trend differences in these placebo outcomes after the HCA, with the exception of an increase in the male share following the HIV/AIDS policy, suggesting that our main results are unlikely to be driven by compositional changes in the patient pool.

FIGURE 3: Total Spending Trends Between Treatment and Control Groups



Note: Figure presents the trends of medical spending around diagnosis of cancer patients by whether the patient is diagnosed before the HCA policy. The dashed lines represent the aggregation window used in the main DiD analysis. Panel A presents treatment and control group trends for the cancer HCA. Panel B presents treatment and control group trends for the HIV/AIDS HCA. Cancers affected by HCA are defined as treated and others as controls. See Appendix Table 1 for the ICD codes used.

Next, we assess the validity of the parallel pre-trends and SUTVA assumptions by plotting raw outcome trends for treated and control groups. Figure 3 presents the trend in total healthcare spending by year relative to the policy for cancer cohorts in Panel A and HIV/AIDS cohorts in Panel B. The solid black line represents the treatment group and the solid gray line represents the control group. The dashed vertical lines represent the year of the policy.

Several patterns are worth noting. First, pre-policy spending trends are similar between treated and control groups for both cancer and HIV/AIDS, consistent with parallel pre-trends. Second, the gap between treated and control spending is visibly larger in the post-policy cohorts than in the pre-policy cohorts. In the pre-policy cohorts, treated cancer patients spend roughly 3.1 million pesos per year ($1,040) more than controls; in the post-policy cohorts, this gap widens to approximately 5.8 million pesos ($1,970). Similarly, HIV/AIDS patients spend 2.3 million pesos per year ($765) more than control diseases before the policy, widening to about 3.3 million pesos ($1,115) afterward. Third, spending patterns among control cancer patients and control diseases for HIV/AIDS are

strikingly stable across the pre- and post-policy cohorts, suggesting that the policy did not spill over to untargeted patients. Control groups experience a slight increase in spending in the post-policy period perhaps reflecting the impact of the COVID-19 pandemic.

Outcomes. We consider two classes of outcomes: healthcare spending and care quality. For spending, we examine total healthcare spending and spending on disease-specific treatments (chemotherapy, radiotherapy, and cancer related surgery for all cancer patients; prescriptions of antiretroviral therapy (ART) for HIV/AIDS; antimycobacterials for tuberculosis; antiepileptics for epilepsy; drugs for obstructive airway diseases for asthma; insulins and blood glucose lowering drugs for diabetes; and antihypertensives, diuretics, beta blockers, calcium channel blockers, and renin-angiotensin agents for hypertension).[7] Appendix Table 2 lists the services and ATC codes used for constructing these outcomes.

For care quality, the cancer analysis uses indicators for hospital admission and receipt of the cancer-specific treatments listed above. The HIV/AIDS analysis uses indicators for vaccination against bacterial and viral diseases (e.g., pneumococcus and flu), and hospital admissions. Appendix Table 2 also lists the services used to identify hospital admissions and vaccinations.

6 Effects of Performance Pay for Cancer

Figure 4 presents results for the cancer HCA; Appendix Figure 3 presents raw outcome trends. Panel A shows that total healthcare spending per patient evolved in parallel across treated and control cancers prior to the policy, then rose by 2.1 million pesos ($717) on average after its introduction, or 27.5% relative to the baseline treated mean of 7.7 million pesos ($2,608). The smaller effects in the first two post-policy years reflect the inclusion of partially treated cohorts, whose outcome windows only partially overlap with the post-policy period; the larger effects in later years reflect fully treated cohorts only.

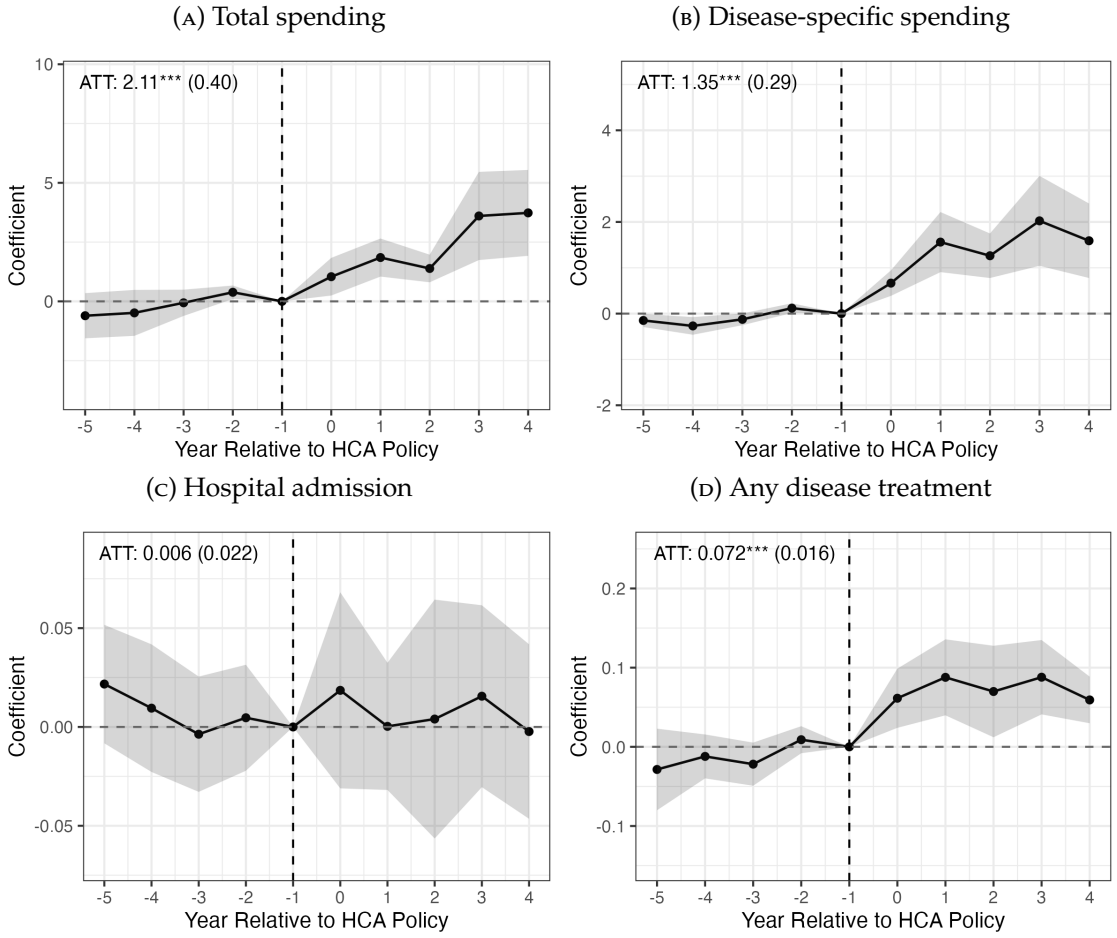
These spending increases are not a mechanical consequence of the performance payments themselves, since our outcome measures spending on healthcare services delivered

[7] We focus on outpatient spending rather than total spending, since we will examine hospital admissions separately in care quality outcomes.

to patients, not insurer revenues. Nor are they attributable to the COVID-19 pandemic, since we identify effects in the years before the pandemic.

Panel B shows that the increase in spending is partly driven by services directly related to cancer management—chemotherapy, radiotherapy, and cancer-related surgeries—which rise by 92.4% relative to the baseline treated mean of 1.5 million pesos ($508).

FIGURE 4: Impact of Cancer HCA on Healthcare Spending and Care Quality



Note: Figure presents coefficients and 95% confidence intervals of the cohort DiD design for the effect of HCA policy on medical utilization of cancer patients (Equation 2). Cancer treatment includes chemotherapy, radiotherapy, and cancer-related surgeries (see Appendix Table 2 for details).. Utilization outcomes are measured in the year of the diagnosis and the two years after, where cancer-induced utilization are most concentrated. The text on the top-left reports the coefficient and standard deviation in parentheses of the effect aggregated in the post-periods. Standard errors are clustered at the ICD-3 code level.

Turning to care quality, Panel C shows no significant change in hospital admissions, suggesting that the spending increases reflect better care rather than deteriorating health. Panel D shows that the rise in disease-specific spending is accompanied by a 7.2 percentage point (p.p.) increase in the likelihood of receiving any cancer-specific treatment, or 12.4%

of the baseline treated mean. Together, these results indicate that performance pay for insurers improved disease management by drawing previously unmanaged patients into treatment, consistent with the prediction that performance pay raises insurer effort on rewarded tasks.

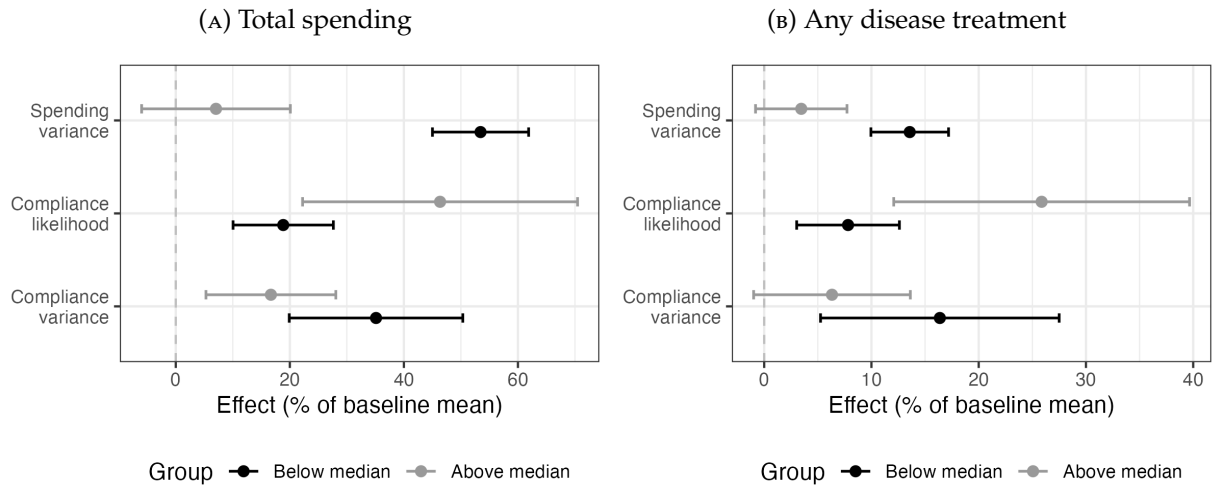
6.1 Test of Predictions

Healthcare spending variance. Our conceptual framework predicts that optimal insurer effort is decreasing in the uncertainty of costs across health conditions (see also Proposition 2(i) in Appendix 1). To test this, we compute the pre-policy variance of healthcare costs for each 3-digit ICD code and split the sample into patients with above- and below-median cost variance (with the median across 3-digit ICD codes weighted by patient counts). The first row of Figure 5 presents the results, expressed as percentages of the baseline treated mean and using as outcome total spending in Panel A and an indicator for having cancer treatments (defined above) in Panel B. Treatment effects are substantially and statistically larger for low-variance conditions. The difference between low- and high-variance groups in total spending equals 1.6 million pesos ($545) or 21% relative to the pre-policy treated mean, and the difference in the likelihood of getting cancer treatments is 5.8 p.p.

Likelihood of meeting quality thresholds. The next prediction is that optimal effort under performance pay is higher the easier it is to meet the quality thresholds and the lower is the variance of the payment (see Proposition 2(ii) in Appendix 1). To test the first prediction we use publicly available information on the quality indicators in 2019, computed using 2018 claims data.[8] Using these pre-policy values, we compute the predicted likelihood of receiving HCA payments for each insurer–3 digit ICD code. This likelihood is the share of quality indicator thresholds that the insurer meets. Specifically,

[8]Measured values of the quality indicators and HCA contributions and payments per insurer-year are available at https://www.minsalud.gov.co/salud/pos/paginas/unidad-de-pago-por-capitacion-upc.aspx, section "Ajuste Ex-Post."

FIGURE 5: Heterogeneity of Cancer HCA



Note: Figure presents coefficients and 95% confidence intervals of a 2x2 cohort difference-in-differences design for the effect of the cancer HCA, estimated separately for patients below and above the median of three measures. Each median is weighted by patient counts. Spending variance is the pre-policy variance of healthcare spending among the patients of each 3-digit ICD code. Compliance likelihood is the predicted compliance $\hat{y}_j$ of the patient's insurer. Compliance variance is the predicted payment variance $\hat{y}_j(1-\hat{y}_j)$ of the patient's insurer. Cancer treatment includes chemotherapy, radiotherapy, and cancer-related surgeries (see Appendix Table 2 for details). Utilization outcomes are averaged over the three years post-diagnosis. Coefficients are expressed as percentages of the baseline treated mean. Standard errors are clustered at the ICD-3 code level.

we approximate equation (1) as:

$$\hat{y}_{\theta j} = \sum_k w_{\theta k} \hat{I}_{\theta k j}$$

where

$$\hat{I}_{\theta k j} = \frac{\mathbf{1}\{\hat{q}_{\theta k j} > q^*_{\theta k}\}}{\sum_j \mathbf{1}\{\hat{q}_{\theta k j} > q^*_{\theta k}\}}$$

and where $\hat{q}_{\theta k j}$ is the value of indicator k for insurer j and 3-digit ICD code θ. Then, we aggregate across health conditions for each insurer weighting by patient counts $\hat{y}_j = \sum_\theta \omega_\theta \hat{y}_{\theta j}$, and split the treatment group into patients whose insurer falls above or below the median of $\hat{y}_j$.

This approach is valid as long as $\hat{y}_{\theta j}$ is a good predictor of both future quality threshold compliance and the magnitude of HCA payments. Appendix Figure 6 provides support for these conditions. Panel A plots $\hat{y}_{\theta j}$ against its 2020 counterpart and shows a strong positive correlation, indicating that an insurer's likelihood of compliance is persistent over time. This persistence is partly mechanical because disease prevalence tends to be stable

across years, but reflects a strong signal of insurer behavior more broadly. Panel B plots $\hat{y}_{\theta j}$ against the net per-capita HCA transfer, defined as payments received minus contributions made ($B_\theta - b_\theta$ from Section 3), divided by the number of patients with disease θ. The strong positive relationship confirms that insurers with a higher predicted likelihood of compliance receive larger net transfers, validating $\hat{y}_{\theta j}$ as a measure of expected financial gain from the HCA.

The second row of Figure 5 presents the results expressed as a percentage of the baseline treated mean. The effect on total healthcare spending in Panel A is 1.7 million pesos ($587) higher among insurers with a high likelihood of compliance relative to their low-predicted compliance counterparts—an effect more than twice as large. The difference between groups is also similar in Panel B for the likelihood of receiving cancer-related treatments.

Performance pay uncertainty. To test the second prediction, that effort decreases with payment uncertainty, the last row in Figure 5 splits the sample by predicted payment variance, defined as $\hat{y}_j(1 - \hat{y}_j)$, above and below the patient-weighted median. For outcomes such as total healthcare spending and cancer-related procedures, insurers with below-median payment variance exhibit treatment effects roughly twice as large as those of their above-median counterparts. For cancer-related procedures in Panel B, the gap between the two groups is 5.8 p.p., or 10% of the baseline treated mean. These findings suggest that whether performance pay translates into better care depends not only on the expected level of payments but also on how confidently insurers can predict receiving them.

6.2 Robustness Checks

We conduct several robustness checks to assess the validity of our cancer HCA results. These exercises are reported in Appendix Figure 7. For each of the outcomes, we report estimates under five alternative specifications. First, we include diagnosis-age-gender fixed effects and insurer-year fixed effects as additional controls in equation (2). Second, we exclude individuals who are ever enrolled with insurers SaludCoop (EPS013) or Cafe-

salud (EPS003), which were terminated in 2015 and 2019, respectively (Serna, 2025). This exercise ensures our results are not driven by care disruptions associated with these forced transitions. Fourth, we exclude individuals who were ever enrolled with insurers Nueva EPS (EPS037), Salud Total (EPS002), and Sanitas (EPS005) which exhibit anomalous jumps in cancer diagnoses in certain years, to verify that our results are not sensitive to these idiosyncratic data patterns. Fifth, we exclude individuals who die during the sample period. Our main results are remarkably robust across these specifications. Finally, Appendix Figure 8 shows that treatment effects are not driven by any single cancer type but are present across all targeted conditions.

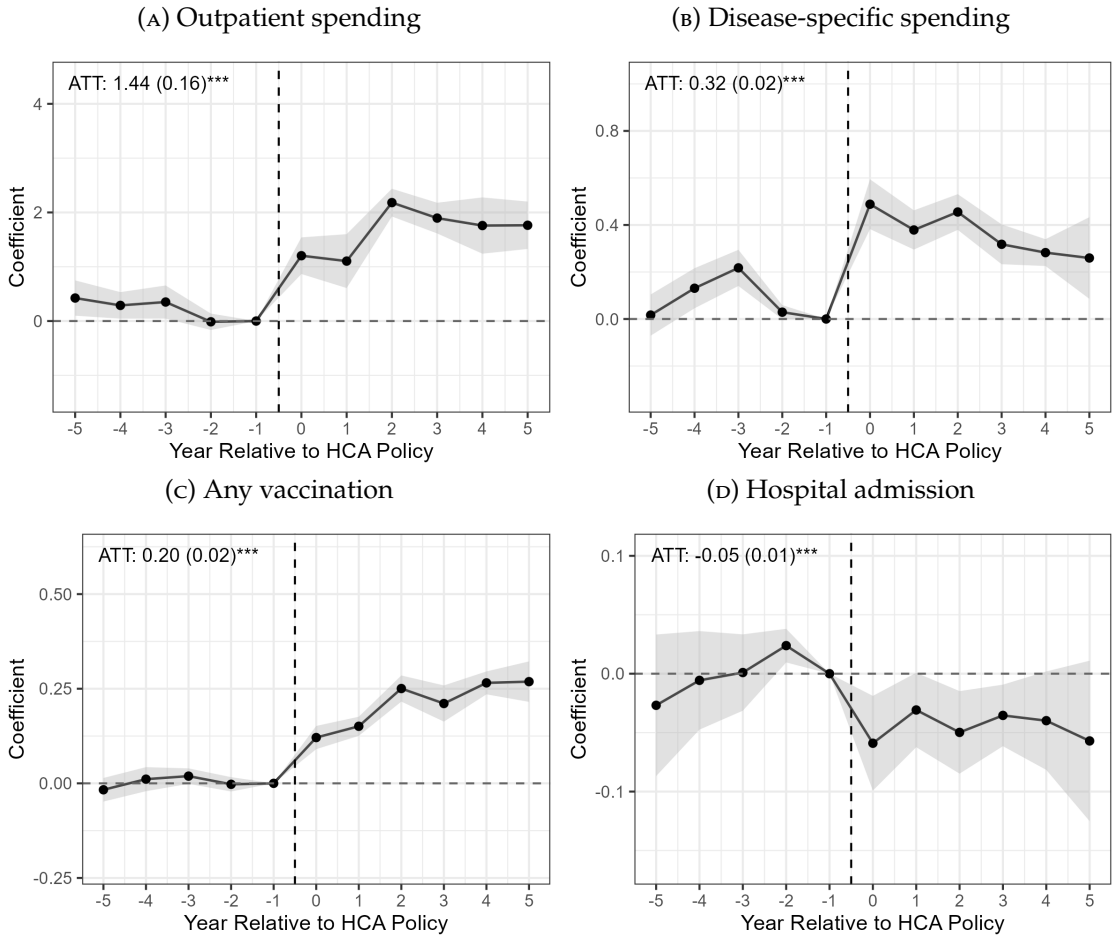
7 Effects of Performance Pay for HIV/AIDS

Figure 6 presents results for the HIV/AIDS policy, with the corresponding raw outcome trends in Appendix Figure 4. Across all outcomes we find evidence of parallel pre-trends, with the exception of disease-specific spending, which drifts modestly upward before the policy, a small movement relative to the sharp post-policy increase (Panel B). Panel A shows that performance pay increased outpatient spending by 1.44 million pesos ($489), or 58.4% relative to baseline. Panel B shows that this increase is driven primarily by ART prescriptions. ART spending increased by nearly 0.32 million pesos ($110) after the policy—almost 10% of the annual per-patient cost of ART in the pre-period.[9]

Because HIV compromises the immune system's ability to fight infection, vaccination against infectious diseases is a cornerstone of appropriate HIV/AIDS management. We find that performance pay increased the likelihood of patients being vaccinated against bacterial and viral infection by 20 p.p. or 157.1% relative to baseline (Panel C). Panel D likewise shows significant reductions in the likelihood of hospital admission (15.5%). Overall, performance pay improved the quality of HIV/AIDS care by protecting against the infections that commonly complicate the disease and reducing the likelihood of experiencing the downstream health consequences of such complications.

[9] See https://hivpreventioncoalition.unaids.org/es/news/colombia-issues-first-ever-compulsory-license-landmark-step-expanding-access-affordable-hiv for details on per-capita ART treatment costs in Colombia.

FIGURE 6: Impact of HIV HCA on Spending and Care Quality



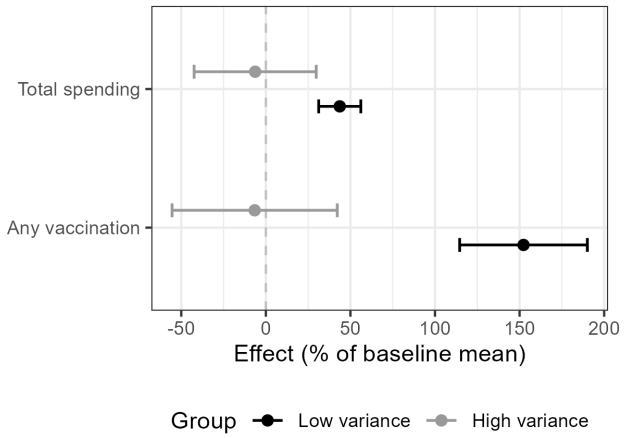
Note: Figure presents results of the cohort event study design for the effect of HCA policy on HIV patients. The text on the top-left reports the coefficient and standard deviation in parentheses of effect on being diagnosed after the policy (Equation 2). Antiretroviral therapy (ART) is defined as claims with an ATC code of J05AF/G/E/J/R.

7.1 Test of Predictions

Unlike the cancer analysis, we cannot reconstruct the insurer-level quality scores for HIV/AIDS because several of the HCA thresholds involve laboratory results, such as viral load measurements, that are not reported in the claims data. However, we can test the prediction that impacts on effort are larger when cost uncertainty is low. We split 3-digit ICD codes into above- and below-median pre-period spending variance (weighted by patient counts) and estimate the event study regression separately within each subsample. Figure 7 presents the results expressed as percentages of the baseline treated mean. The findings confirm that treatment effects across all outcomes are driven by low-variance ICD

codes related to HIV/AIDS. The fact that we obtain qualitatively similar heterogeneity patterns in both the cancer and HIV/AIDS analyses suggests that our results are not an artifact of the particular targeted disease.

FIGURE 7: Heterogeneity of HIV/AIDS HCA by Spending Variance



Note: Figure presents coefficients and 95% confidence intervals of a 2x2 cohort difference-in-differences design for the effect of the HIV/AIDS HCA by variance of healthcare spending above or below the median. The outcomes are total healthcare spending and an indicator for having vaccination against bacterial and viral infections. Spending variance is the pre-policy variance of healthcare spending among the patients of each 3-digit ICD code. Utilization outcomes are averaged over the three years post-diagnosis. Coefficients are expressed as percentages of the baseline treated mean. Standard errors are clustered at the ICD-3 code level.

8 Mechanisms

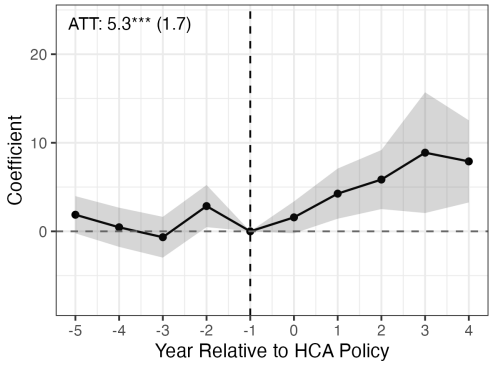
Understanding how insurers translate stronger financial incentives into better patient care is a central question for the design of performance pay in managed care health systems. We investigate mechanisms related to insurers' networks of covered providers, which is the primary dimension of insurer competition in Colombia. Specifically, we use provider characteristics and a measure of wait time as outcomes to examine where patient care is being delivered and authorized by the insurer and how long it takes for the patient to receive it. For provider characteristics we use on the average number of beds during the sample period and for wait time we use an indicator for whether the patient received a disease-specific medication within their year of diagnosis.

For this analysis, we impose the additional sample restriction that patients remain enrolled with the same insurer three years after diagnosis and exclude patients who die

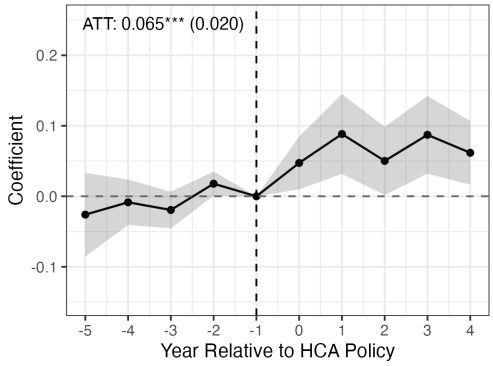
during the sample period. This allows the patient to have sufficient interaction with their insurer's network of providers and avoids potential confounding from insurer switching.

FIGURE 8: Mechanisms for Performance Pay Effects

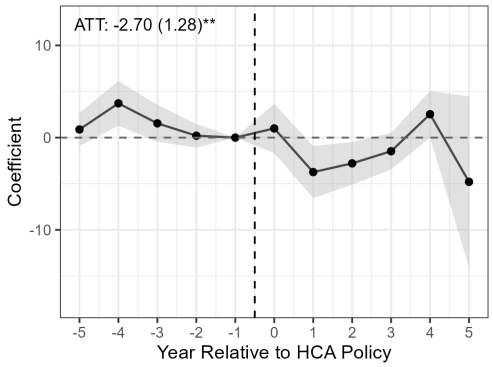
(A) Cancer: Provider Number of Beds



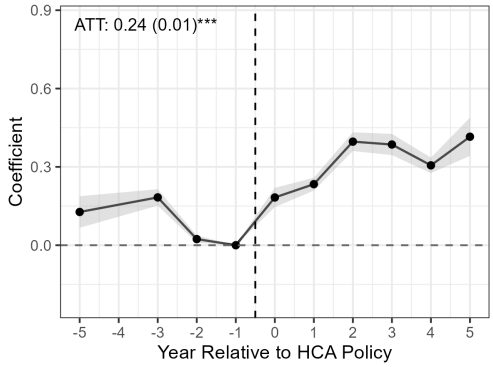
(B) Cancer: Any disease-specific drug within a year



(C) HIV/AIDS: Provider Number of Beds



(D) HIV/AIDS: Any disease-specific drug within a year



Note: Panels (A) and (B) report the effect of HCA policy on the quality of provider visited by cancer patients (see Figure 8 for details). Panels (C) and (D) report the effect of HCA policy on HIV patients (see Figure 6 for details). For each provider, we take the number of inpatient beds from the *Registro Especial de Prestadores de Salud* file from the Ministry of Health and Social Protection. We then calculate for each patient the average measure across visited providers, weighted by the number of claims.

Figure 8, Panel A shows that patients with targeted cancers visited providers with approximately 5.3 more beds on average than patients with untargeted cancers. The corresponding outcome for HIV/AIDS patients (Panel C) shows no change after the introduction of performance pay, perhaps because HIV/AIDS management relies more heavily on timely access to ART medications than on frequent contact with healthcare facilities (NCBI Advisory Council, 2021). Consistent with this, Panels B and D show that receipt of disease-specific treatment within the first year increased by 6.5 p.p. for targeted

cancer patients and by 24 p.p. for HIV/AIDS patients. The cancer effect is similar in magnitude to the effect on receiving any disease-specific treatment within the full three-year post-diagnosis window, whereas the HIV/AIDS effect in the first year alone is twice as large as the effect across all three years combined, suggesting that the policy shortened wait times for medication dispensing among HIV/AIDS patients.

9 Conclusions

This paper studies performance pay for health insurers, combining a theoretical framework with empirical evidence from Colombia. The theory characterizes how quality thresholds, the uncertainty of healthcare costs, and the level and uncertainty of per-capita payments jointly determine whether performance pay induces insurer effort. The theory predicts that insurer effort is decreasing in the uncertainty of costs and per-capita performance payments, but increasing in the ease of meeting the quality thresholds. We test these predictions using individual-level claims data from Colombia between 2011 and 2023, where the government introduced performance pay for HIV/AIDS in 2016 and for cancer in 2019.

We find that performance pay improved disease management for both targeted conditions. For cancer, the policy increased healthcare spending by 2.1 million pesos per patient ($717), driven by a meaningful increase in the utilization of cancer-specific treatments such as chemotherapy, radiotherapy, and cancer-related surgeries. For HIV/AIDS, the policy improved treatment adherence, with a larger share of patients claiming ART medications and receiving preventive vaccinations after its introduction.

Heterogeneity analyses confirm all three theoretical predictions: treatment effects are more than twice as large among insurers with a higher ex-ante likelihood of receiving payments, consistent with effort increasing in the ease of meeting quality thresholds; twice as large among insurers with below-median payment variance, consistent with effort decreasing in the uncertainty of being paid; and more than three times as large for health conditions with below-median cost variance, consistent with risk-averse insurers concentrating effort on more homogeneous conditions. To achieve these gains, insurers

directed cancer patients to larger, more specialized facilities within their networks and shortened wait times for medication dispensing for HIV/AIDS patients.

These findings have important implications for the design of performance pay in healthcare markets. Our results suggest that targeting payments to specific, verifiable conditions is a key determinant of the policy's effectiveness.

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Appendix 1 Theoretical Framework

1.1 Model

We model a health insurance market in which premiums are fixed and an insurer chooses a vector of efforts $\mathbf{e}_\theta = (e_{1\theta}, \ldots, e_{K\theta})$, where $e_{k\theta} \geq 0$ denotes effort along quality dimension k for consumer type $\theta \in \{0, 1, \ldots, I\} \equiv \Theta$, with $\theta = 0$ denoting healthy individuals. Each dimension k corresponds to a measurable quality indicator, such as ART prescription rates, vaccination rates, or early disease detection, which the regulator can contract on.

Assumption 1 (Demand and costs)**.** *Insurer demand from consumer type θ, $s_\theta(\mathbf{e}_\theta)$, depends only on effort directed at type θ and satisfies $\frac{\partial s_\theta}{\partial e_{k\theta}} > 0$, $\frac{\partial^2 s_\theta}{\partial e_{k\theta}^2} < 0$, and $\frac{\partial^2 s_\theta}{\partial e_{k\theta}\partial e_{k'\theta}} = 0$ for all $k \neq k'$. Cost per type-θ enrollee is:*

$$C_\theta = c_\theta(\mathbf{e}_\theta)\,(1 + \eta_\theta), \qquad \mathbb{E}[\eta_\theta] = 0, \quad \mathrm{Var}(\eta_\theta) = \sigma_\theta^2,$$

so that $\mathbb{E}[C_\theta] = c_\theta(\mathbf{e}_\theta)$ and $\mathrm{Var}(C_\theta) = \sigma_\theta^2\, c_\theta(\mathbf{e}_\theta)^2$, where c_θ satisfies $\frac{\partial c_\theta}{\partial e_{k\theta}} > 0$, $\frac{\partial^2 c_\theta}{\partial e_{k\theta}^2} > 0$, and $\frac{\partial^2 c_\theta}{\partial e_{k\theta}\partial e_{k'\theta}} = 0$ for all $k \neq k'$.

Assumption 1 states that consumers prefer insurers that invest more in each dimension of care quality and that higher effort along any dimension raises costs. Cost uncertainty is multiplicative, meaning that the dispersion of realized costs scales with their expected level, which is a common feature of healthcare costs (Brown et al., 2014).

Assumption 2 (Performance payments)**.** *The regulator measures the insurer's quality on K indicators. The measured score on indicator k is a noisy signal of effort, $q_{k\theta} = \tau_{k\theta}(e_{k\theta}) + \varepsilon_{k\theta}$, with $\varepsilon_{k\theta} \sim \mathcal{N}(0, \sigma_k^2)$ independent across k and independent of η_θ, and where $\tau_{k\theta}(\cdot)$ is increasing in effort. For each indicator the regulator sets a threshold $q^*_{k\theta}$ and pays the insurer in proportion to the distance between the measured score and the threshold, with weights $w_k \geq 0$, $\sum_k w_k = 1$. The per-capita performance payment is:*

$$B_\theta = p_\theta \sum_k w_k \max\{q_{k\theta} - q^*_{k\theta},\, 0\}. \tag{3}$$

Define the expected distance to the threshold, its standardized value, and the compliance likelihood as:

$$d_{k\theta} \equiv \tau_{k\theta}(e_{k\theta}) - q^*_{k\theta}, \qquad z_{k\theta} \equiv \frac{d_{k\theta}}{\sigma_k}, \qquad \Phi_{k\theta} \equiv \Phi(z_{k\theta}) = \Pr\left(q_{k\theta} > q^*_{k\theta}\right),$$

where Φ and ϕ denote the standard normal cdf and pdf. Because equation (3) represents a censored normal random variable, the payment has mean and variance

$$\mathbb{E}[B_\theta] = p_\theta \sum_k w_k \, m_{k\theta}, \qquad \text{Var}(B_\theta) = p_\theta^2 \sum_k w_k^2 \, v_{k\theta}, \tag{4}$$

where

$$\begin{aligned} m_{k\theta} &= d_{k\theta}\,\Phi_{k\theta} + \sigma_k\,\phi(z_{k\theta}), \\ v_{k\theta} &= d_{k\theta}^2\,\Phi_{k\theta}(1-\Phi_{k\theta}) + \sigma_k d_{k\theta}\,\phi(z_{k\theta})\big(1-2\Phi_{k\theta}\big) + \sigma_k^2\big(\Phi_{k\theta} - \phi(z_{k\theta})^2\big). \end{aligned} \tag{5}$$

Under Assumption 2, payment uncertainty arises because the insurer does not know how its effort will translate into the measured score (σ_k) and thus clearing the threshold is uncertain. Performance payments in this case are a kinked function of effort.

In the limit where $q^*_{k\theta} \to -\infty$, then $\Phi_{k\theta} \to 1$, $m_{k\theta} \to d_{k\theta}$, and $v_{k\theta} \to \sigma_k^2$, and the piece-wise performance pay contract collapses to a linear contract of the following form: $B_\theta = p_\theta \sum_k w_k q_{k\theta}$.

Assumption 3 (Insurer's problem)**.** *The insurer's profit function is:*

$$\Pi = \sum_{\theta\in\Theta} \left\{\mathbb{E}[\pi_\theta] - \frac{\rho}{2}\,\text{Var}(\pi_\theta)\right\}, \qquad \pi_\theta = \left(R_\theta - C_\theta + B_\theta\right) s_\theta(\mathbf{e}_\theta),$$

where R_θ is the risk-adjusted per-capita payment from the government and $\rho > 0$ is a risk-aversion parameter.

Using Assumptions 1–2 and the independence of C_θ and B_θ, the insurer's objective

function is

$$\Pi = \sum_{\theta} \left\{ \left[R_\theta - c_\theta + p_\theta \sum_k w_k m_{k\theta} \right] s_\theta - \frac{\rho}{2} s_\theta^2 \left[\sigma_\theta^2 c_\theta^2 + p_\theta^2 \sum_k w_k^2 v_{k\theta} \right] \right\}.$$

First-order condition. Let $\tau'_{k\theta} \equiv \partial \tau_{k\theta} / \partial e_{k\theta}$, the following identities simplify the derivatives with respect to effort (see Appendix 1):

$$\frac{\partial m_{k\theta}}{\partial e_{k\theta}} = \tau'_{k\theta}\, \Phi_{k\theta}, \qquad \frac{\partial v_{k\theta}}{\partial e_{k\theta}} = 2\, \tau'_{k\theta}\, m_{k\theta} \big(1 - \Phi_{k\theta}\big) \;\geq\; 0. \tag{6}$$

Given separability, the first-order condition (FOC) with respect to $e_{k\theta}$ is

$$\begin{aligned} FOC(e_{k\theta}) = &\left[R_\theta - c_\theta + p_\theta \textstyle\sum_{k'} w_{k'} m_{k'\theta} \right] \frac{\partial s_\theta}{\partial e_{k\theta}} - \left[\frac{\partial c_\theta}{\partial e_{k\theta}} - p_\theta w_k \tau'_{k\theta} \Phi_{k\theta} \right] s_\theta \\ &- \rho \left[\sigma_\theta^2 c_\theta \frac{\partial c_\theta}{\partial e_{k\theta}} + p_\theta^2 w_k^2\, \tau'_{k\theta}\, m_{k\theta} (1 - \Phi_{k\theta}) \right] s_\theta^2 - \rho \left[\sigma_\theta^2 c_\theta^2 + p_\theta^2 \textstyle\sum_{k'} w_{k'}^2 v_{k'\theta} \right] s_\theta \frac{\partial s_\theta}{\partial e_{k\theta}} = 0. \end{aligned} \tag{7}$$

The first line of equation (7) is the risk-neutral problem (i.e., $\rho = 0$). In this problem, performance pay reduces the effective marginal cost of effort from $\frac{\partial c_\theta}{\partial e_{k\theta}}$ to $\frac{\partial c_\theta}{\partial e_{k\theta}} - p_\theta w_k \tau'_{k\theta} \Phi_{k\theta}$. The second line is the risk premium or how much the insurer would be willing to pay to avoid risk. Effort raises this profit variance through three channels. First, it raises expected costs and hence their dispersion. Second, it raises the variance of the performance payment, since when effort is very low the payment almost surely is zero but close to the threshold the payment becomes uncertain. Third, it raises enrollment, which scales the exposure to both sources of risk.

Proposition 1 (Risk-neutral benchmark)**.** *Suppose $\rho = 0$. Optimal effort $e^*_{k\theta}$ is (i) increasing in the per-capita payment p_θ; (ii) increasing in the indicator weight w_k; (iii) increasing in the compliance likelihood $\Phi_{k\theta}$, i.e., decreasing in the threshold $q^*_{k\theta}$; (iv) decreasing in measurement noise σ_k if and only if*

$$z_{k\theta}\, \tau'_{k\theta}\, s_\theta \;>\; \sigma_k \frac{\partial s_\theta}{\partial e_{k\theta}},$$

which requires $d_{k\theta} > 0$; and (v) independent of cost variance σ_θ^2.

Proposition 1 largely follows Lazear and Rosen (1981). Part (iv) specifically states that

measurement noise lowers the probability that marginal effort is rewarded (i.e., $\tau'_{k\theta}$ is multiplied by $1/\sigma_k$), reducing the return to effort even for a risk-neutral agent. But when the insurer expects to fall short of clearing the threshold (i.e., $z < 0$), more noise raises the chance of a favorable draw and increases effort.

Proposition 2 (Risk aversion)**.** *Suppose $\rho > 0$. Optimal effort $e^*_{k\theta}$ is (i) decreasing in cost variance σ^2_θ; (ii) decreasing in payment variance $v_{k\theta}$ holding $\Phi_{k\theta}$ fixed; (iii) increasing in p_θ and w_k provided ρ is sufficiently small.*

Proposition 2 indicates that the insurer's optimal effort under risk aversion largely satisfies the conditions of the risk neutral problem with the additional result that optimal effort will be higher for tasks that lead to more certain expected profits and for health conditions with less dispersion in costs across individuals.

Proposition 3 (Multitasking)**.** *An increase in the weight w_k placed on indicator k raises optimal effort $e^*_{k\theta}$ along dimension k but has no effect on optimal effort $e^*_{k'\theta}$ along any other dimension $k' \neq k$.*

Proposition 3 relies on the assumption that demand and costs are independent across effort dimensions and on that fact that thresholds are indicator-specific, so that $\Phi_{k'\theta}$ and $v_{k'\theta}$ do not depend on $e_{k\theta}$. Empirically, if performance pay diverts effort away from unrewarded health conditions, then comparing outcomes between targeted and untargeted conditions would violate the Stable Unit Treatment Value Assumption (SUTVA).

Proposition 4 (Disease targeting)**.** *Consider a pooled scheme in which indicator k is aggregated across diseases, $\tau_k = \sum_\theta \omega_\theta \tau_{k\theta}$ with population weights ω_θ, a single threshold q^*_k, and a single payment rate p. Optimal effort $e^*_{k\theta}$ is lower under pooling than under disease-specific payments whenever*

$$\omega_\theta \, p \, \Phi\big(z_k^{\text{pool}}\big) \;<\; p_\theta \, \Phi_{k\theta}.$$

Proposition 4 suggests that under pooling, effort directed at disease θ moves the aggregate score only in proportion to ω_θ, so the insurer captures a fraction of the return it would earn under a disease-specific payment. Pooling also changes the compliance likelihood. Averaging across diseases reduces noise in the aggregate score, which can

raise Φ, but it also means that the threshold is cleared or missed for all diseases at once. Also note that if the regulator scales up the pooled payment, or pooling raises compliance likelihood enough, it could counteract the effect of ω_θ and lead to higher effort. This means that pooled payments are not always detrimental for welfare, but the condition outlined in Proposition 4 can help explain why pooled payments such as those in Medicare Advantage's star rating system have failed to improve patient outcomes.

Proposition 5 (Risk selection)**.** *Let*

$$\Lambda_\theta \equiv \frac{\partial \Pi}{\partial s_\theta} = R_\theta - c_\theta + p_\theta \sum_k w_k m_{k\theta} - \rho\, s_\theta \Big[\sigma_\theta^2 c_\theta^2 + p_\theta^2 \sum_k w_k^2 v_{k\theta}\Big].$$

Then (i) the insurer invests in attracting type θ if $\Lambda_\theta > 0$ and in deterring it if $\Lambda_\theta < 0$; (ii) under perfect risk adjustment, $R_\theta = c_\theta$, a risk-averse insurer strictly prefers lower-cost types; (iii) performance payments raise the relative attractiveness of types with a larger expected distance to the threshold $d_{k\theta}$ and, when $\rho > 0$, lower payment variance $v_{k\theta}$.

Proposition 5 establishes that another margin along which insurers may respond to performance pay is risk selection (Fioretti and Wang, 2023). Part (ii) implies that even with perfect risk adjustment a risk-averse insurer will prefer to enroll lower-cost enrollees because cost variance scales with the expected cost. Part (iii) shows that because performance payments reward measured quality, insurers gain from enrolling patients on whom high scores are easier to achieve.

1.2 Proofs

Moments of the censored normal. Fix k, θ and drop subscripts. Let $Y = d + \varepsilon$ with $\varepsilon \sim \mathcal{N}(0, \sigma^2)$, $z = d/\sigma$, and $X = \max\{Y, 0\}$. Substituting $Y = d + \sigma u$ with $u \sim \mathcal{N}(0,1)$ yields the following first moment:

$$\mathbb{E}[X] = \int_{-z}^{\infty} (d + \sigma u)\, \phi(u)\, du = d\, \Phi(z) + \sigma\, \phi(z),$$

using $\int_a^\infty u\, \phi(u)\, du = \phi(a)$ and $\phi(-z) = \phi(z)$.

For the second moment, we have:

$$\mathbb{E}[X^2] = \int_{-z}^{\infty} (d+\sigma u)^2 \phi(u)\, du = d^2\Phi(z) + 2d\sigma\, \phi(z) + \sigma^2\big(\Phi(z) - z\,\phi(z)\big) = (d^2+\sigma^2)\Phi(z) + d\sigma\, \phi(z),$$

using integration by parts. Then $\mathrm{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2$, which expands to (5). Since the ε_k are independent, $\mathrm{Var}(B) = p^2 \sum_k w_k^2 \,\mathrm{Var}(X_k)$, giving (4).

Identities (6). Effort on indicator k enters the payment moments only through $d_{k\theta} = \tau_{k\theta}(e_{k\theta}) - q^*_{k\theta}$, so by the chain rule

$$\frac{\partial m_{k\theta}}{\partial e_{k\theta}} = \frac{\partial m_{k\theta}}{\partial d_{k\theta}}\, \tau'_{k\theta} = \Phi_{k\theta}\, \tau'_{k\theta}, \qquad \frac{\partial v_{k\theta}}{\partial e_{k\theta}} = \frac{\partial v_{k\theta}}{\partial d_{k\theta}}\, \tau'_{k\theta} = 2\, m_{k\theta}\big(1 - \Phi_{k\theta}\big)\, \tau'_{k\theta},$$

where $\partial m_{k\theta}/\partial d_{k\theta} = \Phi_{k\theta}$ and $\partial v_{k\theta}/\partial d_{k\theta} = 2m_{k\theta}(1 - \Phi_{k\theta})$.

Proof of Proposition 1. Set $\rho = 0$ in (7). Strict concavity of s_θ and convexity of c_θ ensure the second-order condition holds, so by the implicit function theorem, we have:

$$\begin{aligned}
\frac{\partial FOC}{\partial p_\theta} &= \Big(\sum_{k'} w_{k'} m_{k'\theta}\Big)\frac{\partial s_\theta}{\partial e_{k\theta}} + w_k \tau'_{k\theta}\Phi_{k\theta} s_\theta > 0,\\
\frac{\partial FOC}{\partial w_k} &= p_\theta m_{k\theta}\frac{\partial s_\theta}{\partial e_{k\theta}} + p_\theta \tau'_{k\theta}\Phi_{k\theta} s_\theta > 0,\\
\frac{\partial FOC}{\partial(-q^*_{k\theta})} &= p_\theta w_k\left[\Phi_{k\theta}\frac{\partial s_\theta}{\partial e_{k\theta}} + \tau'_{k\theta}\frac{\phi(z_{k\theta})}{\sigma_k} s_\theta\right] > 0,\\
\frac{\partial FOC}{\partial \sigma_k} &= p_\theta w_k\, \phi(z_{k\theta})\left[\frac{\partial s_\theta}{\partial e_{k\theta}} - \frac{z_{k\theta}\tau'_{k\theta}}{\sigma_k} s_\theta\right],
\end{aligned}$$

establishing (i)–(iv). Part (v) holds because σ_θ^2 enters (7) only through terms multiplied by ρ. □

Proof of Proposition 2. For $\rho > 0$,

$$\frac{\partial FOC}{\partial \sigma_\theta^2} = -\rho\left[c_\theta \frac{\partial c_\theta}{\partial e_{k\theta}} s_\theta^2 + c_\theta^2 s_\theta \frac{\partial s_\theta}{\partial e_{k\theta}}\right] < 0,$$

establishing (i). Holding $\Phi_{k\theta}$ fixed, $v_{k\theta}$ enters (7) only through $-\rho p_\theta^2 w_k^2 v_{k\theta}\, s_\theta \frac{\partial s_\theta}{\partial e_{k\theta}} < 0$, establishing (ii). For (iii),

$$\frac{\partial FOC}{\partial p_\theta} = \underbrace{\Big(\sum_{k'} w_{k'} m_{k'\theta}\Big)\frac{\partial s_\theta}{\partial e_{k\theta}} + w_k \tau'_{k\theta}\Phi_{k\theta} s_\theta}_{>0} - 2\rho p_\theta \Big[w_k^2 \tau'_{k\theta} m_{k\theta}(1-\Phi_{k\theta}) s_\theta^2 + \sum_{k'} w_{k'}^2 v_{k'\theta}\, s_\theta \frac{\partial s_\theta}{\partial e_{k\theta}}\Big],$$

which is positive for ρ below a finite threshold and negative above it. The argument for w_k is analogous. □

Proof of Proposition 3. By the zero cross-partial assumptions on demand and costs and because $m_{k'\theta}$, $v_{k'\theta}$, and $\Phi_{k'\theta}$ depend only on $e_{k'\theta}$, $FOC(e_{k'\theta})$ does not depend on w_k for $k' \neq k$. Hence $\partial e^*_{k'\theta}/\partial w_k = 0$. The effect on $e^*_{k\theta}$ follows from Proposition 1(ii) or 2(iii). □

Proof of Proposition 4. Under pooling the marginal return to effort on indicator k in the first line of (7) becomes $\omega_\theta\, \tau'_{k\theta}\, w_k\, p\, \Phi(z_k^{\text{pool}})$, versus $\tau'_{k\theta}\, w_k\, p_\theta\, \Phi_{k\theta}$ under disease-specific payments. The result follows from the implicit function theorem whenever the former is smaller. □

Proof of Proposition 5. By Assumption 1, s_θ depends only on $\mathbf{e}_\theta$, so effort is the insurer's only instrument for changing enrollment of type θ. The contribution of the enrollment margin to $FOC(e_{k\theta})$ is $\Lambda_\theta\, \partial s_\theta/\partial e_{k\theta}$; since $\partial s_\theta/\partial e_{k\theta} > 0$, this raises optimal effort when $\Lambda_\theta > 0$ and lowers it when $\Lambda_\theta < 0$, establishing (i). Under $R_\theta = c_\theta$, $\partial\Lambda_\theta/\partial c_\theta = -2\rho s_\theta \sigma_\theta^2 c_\theta < 0$ for $\rho > 0$, establishing (ii). Finally, $\partial\Lambda_\theta/\partial d_{k\theta} = p_\theta w_k \Phi_{k\theta} - 2\rho s_\theta p_\theta^2 w_k^2 m_{k\theta}(1-\Phi_{k\theta})$, whose first term is positive, and $\partial\Lambda_\theta/\partial v_{k\theta} = -\rho s_\theta p_\theta^2 w_k^2 < 0$, establishing (iii). □

Appendix 2 Data Cleaning Details

Cancer sample. The main cancer analysis uses a cross-sectional person-level dataset in which each observation is a cancer patient's first three years of care following diagnosis.

The data sources are the *Base de Suficiencia* (individual health records) from 2013–2023, provided by each insurer. Insurers that appear in the health claims data are those that pass the Ministry of Health's data quality filters. Enrollment records supply demographics information such as age and sex.

We assign each person their earliest cancer diagnosis (if any) using three-digit ICD-10 codes. A cancer diagnosis is recognized only if the person accumulates more than one claim with that diagnosis code that is neither a prescription nor a consultation service, across the full sample period. See Appendix Table 1 for the ICD codes that define a cancer diagnosis.

The sample is then restricted to individuals with a single cancer diagnosis, non-missing sex and age at diagnosis, no concurrent HIV diagnosis, and an insurer whose claims passed the data quality filters in all three follow-up years. Individuals with zero total spending over the window are dropped (this dropped fewer than five observations). Insurer–diagnosis–cohort cells whose patient count or mean spending exceeds three times the average of all other years for that insurer–diagnosis combination are excluded (see Appendix Figure 9). Because the HCA policy took effect in 2019 and the follow-up window is two years, the first treated diagnosis cohort is 2017.

Outcome variables. Spending outcomes cover total spending and itemized categories (outpatient, prescriptions, surgery, chemotherapy, radiotherapy), all summed over the three-year follow-up window. We additionally construct hospital admission counts and average provider-level characteristics (number of beds and ICU availability) weighted by the number of claims. Provider level characteristics are obtained from December snapshots from 2011 to 2016 of the *Registro Especial de Prestadores de Salud* available at https://prestadores.minsalud.gov.co/habilitacion/.

HIV sample. The HIV analysis uses the same data source and diagnosis identification procedure, but differs in several respects. The outcome window covers only the diagnosis year itself rather than three subsequent years. Unlike cancer, the diagnosis threshold counts all claims bearing the HIV diagnosis code. Additional outcomes include ART

uptake, vaccines, and HIV complications. Individuals with a concurrent cancer diagnosis are excluded, as are those ever enrolled in two insurers (EPS013 and EPS003) that were terminated in 2015 and 2019, respectively. The same outlier cell exclusion applies.

Appendix Table 1: ICD Code List for Treated and Control Diseases

Disease	ICD-3 Codes
Panel A: Cancer	
Treated	
Breast Cancer	C50
Cervical Cancer	C53
Colorectal Cancer	C18, C19, C20, C21
Gastric Cancer	C16
Lung Cancer	C33, C34
Lymphoma & Leukemia	C82, C83, C84, C85, C91, C92
Prostate Cancer	C61
Control	
Bone Cancer	C40, C41
CNS Cancer	C69, C70, C71, C72
Endocrine Cancer	C73, C74, C75
Oral & Pharyngeal Cancer	C00, C01, C02, C03, C04, C05, C06, C07, C08, C09, C10, C11, C12, C13, C14
Other Digestive Cancer	C15, C17, C22, C23, C24, C25, C26
Other Female Genital Cancer	C51, C52, C54, C55, C56, C57, C58
Other Hematologic Cancer	C81, C86, C87, C88, C89, C90, C93, C94, C95, C96
Other Male Genital Cancer	C60, C62, C63
Other Respiratory Cancer	C30, C31, C32, C35, C36, C37, C38, C39
Skin Cancer	C43, C44
Soft Tissue Cancer	C45, C46, C47, C48, C49
Urinary Tract Cancer	C64, C65, C66, C67, C68
Panel B: HIV/AIDS	
Treated	
HIV	B20, B21, B22, B23, B24
Control	
Asthma	J45, J46, Y55
Diabetes	E10, E11, E12, E13, E14, E15, G59, G63, G99, H28, H36, M14, N08, T38, Y42, Z13, Z83
Epilepsy	G40, G41, R56, T42, Y46
Hypertension	I10, I11, I12, I13, I15, I27, I67
Tuberculosis	A15, A16, A17, A18, A19, B90, J65, M01, N74, Z03, Z11, Z20, Z23

Note: This table present the ICD codes used to identify cancer and HIV patients in the main sample. ICD codes are shorten to the 3-digits level for brevity.

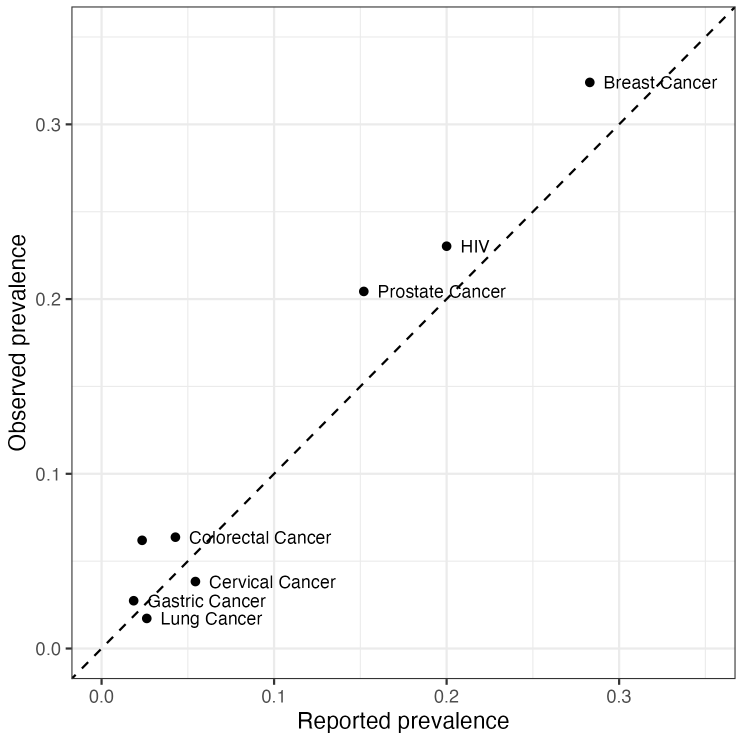
Appendix Table 2: Outcome Definitions for Cancer and Chronic Condition Patients

Variable	Category	Definition
Cancer Treatment	Chemotherapy	Induction chemotherapy; intrathecal chemotherapy; mono-chemotherapy (single cycle); poly-chemotherapy (low and high risk); intra-arterial and intracavitary administration.
Cancer Treatment	Radiotherapy	Teletherapy with radioisotopes, photons, or electrons; brachytherapy; conformal radiotherapy; stereotactic radiosurgery with linear accelerator.
Cancer Treatment	Surgery	Excision, resection, and ablation procedures from the surgical section of CUPS by cancer site: breast; male and female genital organs; digestive organs; respiratory tract; urinary tract, endocrine glands, hematologic system, CNS, skin, and oral/pharyngeal cavity. Excludes cosmetic, reconstructive, transplant, and non-oncologic procedures.
Admission	Admission	Low, medium, and high complexity inpatient stays; intensive care; intermediate care; partial hospitalization.
Vaccine	Vaccine	Prophylactic vaccinations against bacterial and viral diseases; antitoxin administration (tetanus, botulism, diphtheria).
Chronic Disease Drug	ART (HIV/AIDS)	Antiretroviral therapy: nucleoside and nucleotide reverse transcriptase inhibitors, non-nucleoside reverse transcriptase inhibitors, protease inhibitors, integrase inhibitors, and fixed-dose antiviral combinations (ATC J05AE, J05AF, J05AG, J05AJ, J05AR).
Chronic Disease Drug	Antimycobacterials (Tuberculosis)	Drugs for treatment of tuberculosis (ATC J04A).
Chronic Disease Drug	Antiepileptics (Epilepsy)	Antiepileptics (ATC N03A).
Chronic Disease Drug	Obstructive airway drugs (Asthma)	Drugs for obstructive airway diseases: inhaled and systemic adrenergics, glucocorticoids, anticholinergics, and other agents (ATC R03).
Chronic Disease Drug	Diabetes drugs (Diabetes)	Insulins and blood glucose lowering drugs (ATC A10).
Chronic Disease Drug	Antihypertensives (Hypertension)	Antihypertensives, diuretics, beta blocking agents, calcium channel blockers, and agents acting on the renin-angiotensin system (ATC C02, C03, C07, C08, C09).

Note: This table present the description of the service and ATC codes used to identify medical utilization outcomes in the main sample.

Appendix 3 Additional Descriptives and Robustness Checks

Appendix Figure 1: Comparison of Prevalence



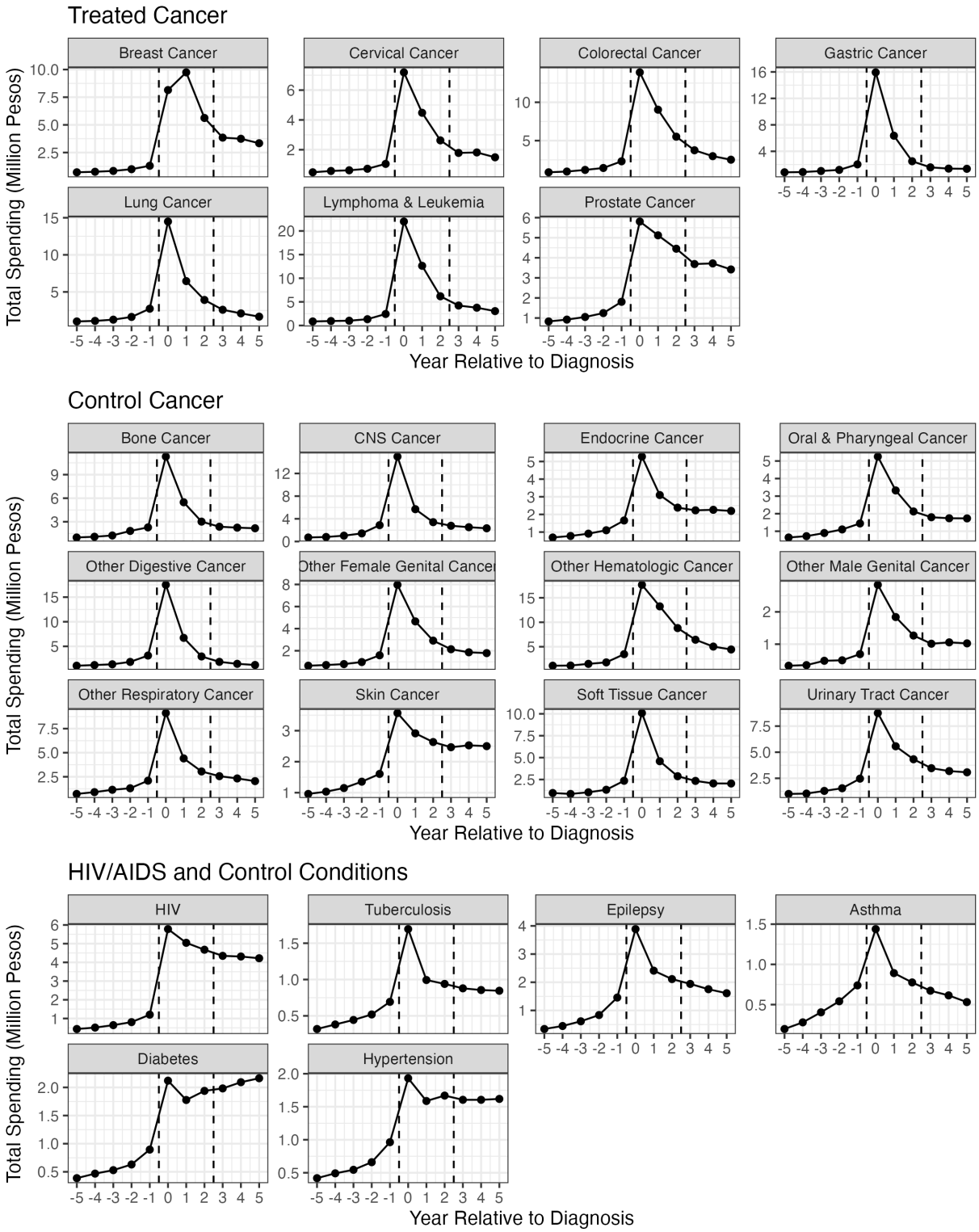
Note: Figure compares the prevalence observed in the sample against the official statistics reported by the HCA in 2016. Observed prevalence is defined as the ratio between patients with any diagnosing claims associated with the cancer, and the total number of enrollees in percentage points. Reported prevalence are obtained by official reports published by the HCA. Only prevalence of the treated diseases are reported.

Appendix Table 3: Summary Statistics of Treated and Control Diseases

Panel A: Cancer		
	Treated Cancer	Control Cancer
Male	0.407 (0.491)	0.404 (0.491)
Age at diagnosis	58.5 (17.8)	53.9 (20.7)
Pre-diagnosis spending	1.30 (5.26)	1.44 (5.59)
Total spending	6.03 (10.1)	3.89 (7.41)
Disease-specific spending	1.13 (2.91)	0.463 (1.76)
Any disease treatment	0.574 (0.494)	0.512 (0.500)
Any admission	0.504 (0.500)	0.432 (0.495)
Patients	138,662	154,593
Unique ICD codes	16	66
Panel B: HIV/AIDS		
	HIV/AIDS	Synthetic Control
Male	0.746 (0.435)	0.507 (0.500)
Age at diagnosis	36.6 (15.1)	26.8 (24.0)
Pre-diagnosis spending	1.03 (4.63)	0.867 (4.49)
Total spending	4.31 (6.24)	1.83 (5.62)
Disease-specific spending	0.383 (1.08)	0.0476 (0.482)
Any vaccination	0.125 (0.331)	0.0792 (0.270)
Any admission	0.304 (0.460)	0.323 (0.468)
Patients	8,474	86,485
Unique ICD codes	5	32

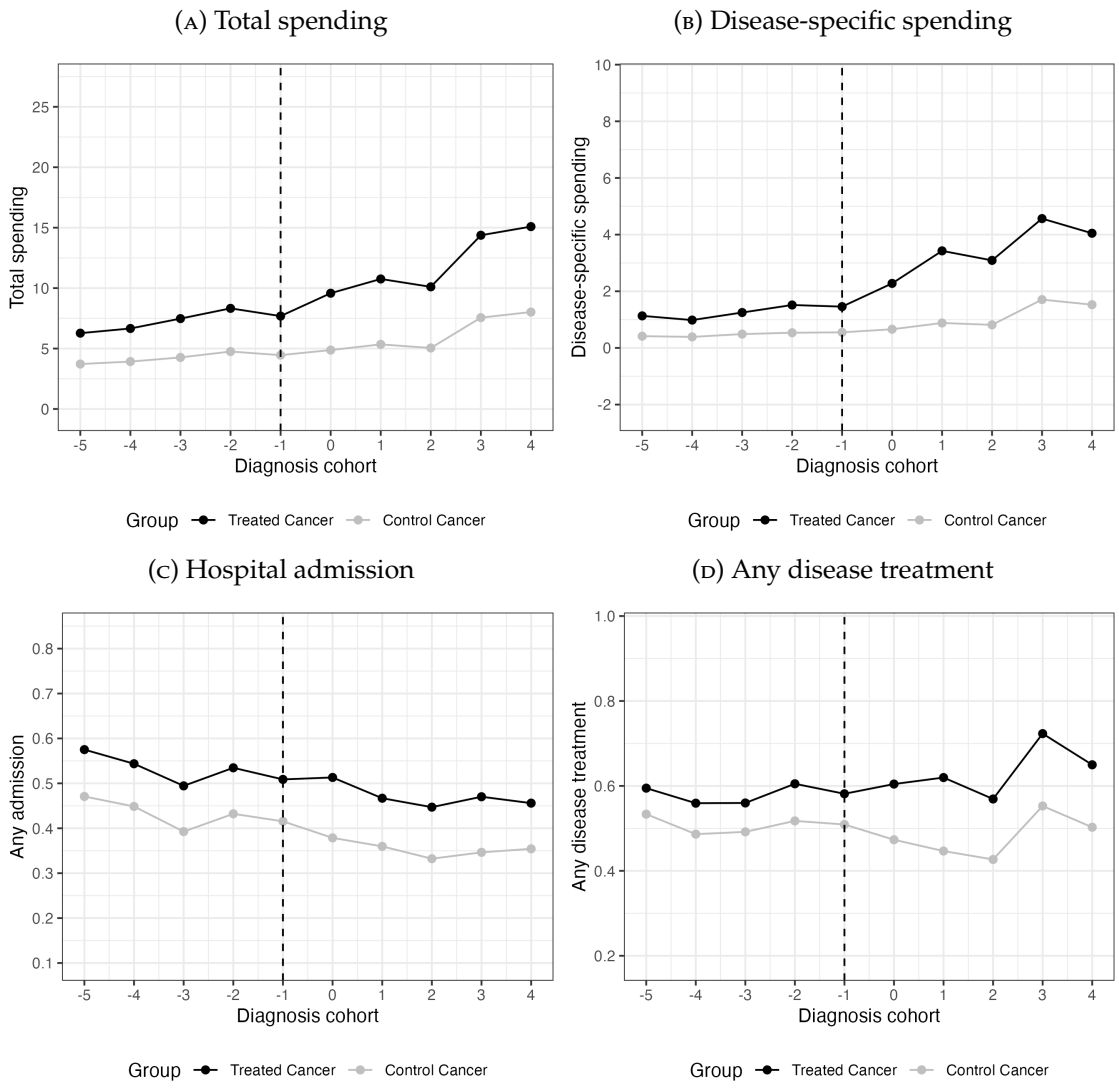
Note: Table presents means and standard deviations in parentheses for the cohorts diagnosed before the policy: before 2017 for cancer and before 2014 for HIV/AIDS. Cancers affected by HCA are defined as treated and others as controls. See Appendix Table 1 for the ICD codes used. The synthetic control weights each control condition by its synthetic control weight. Pre-diagnosis spending is the total spending in the year before the diagnosis. Spending is in million pesos per year. Utilization outcomes are measured in the year of the diagnosis and the two years after, where cancer-induced utilization are most concentrated.

Appendix Figure 2: Healthcare Spending by Year Relative to Diagnosis



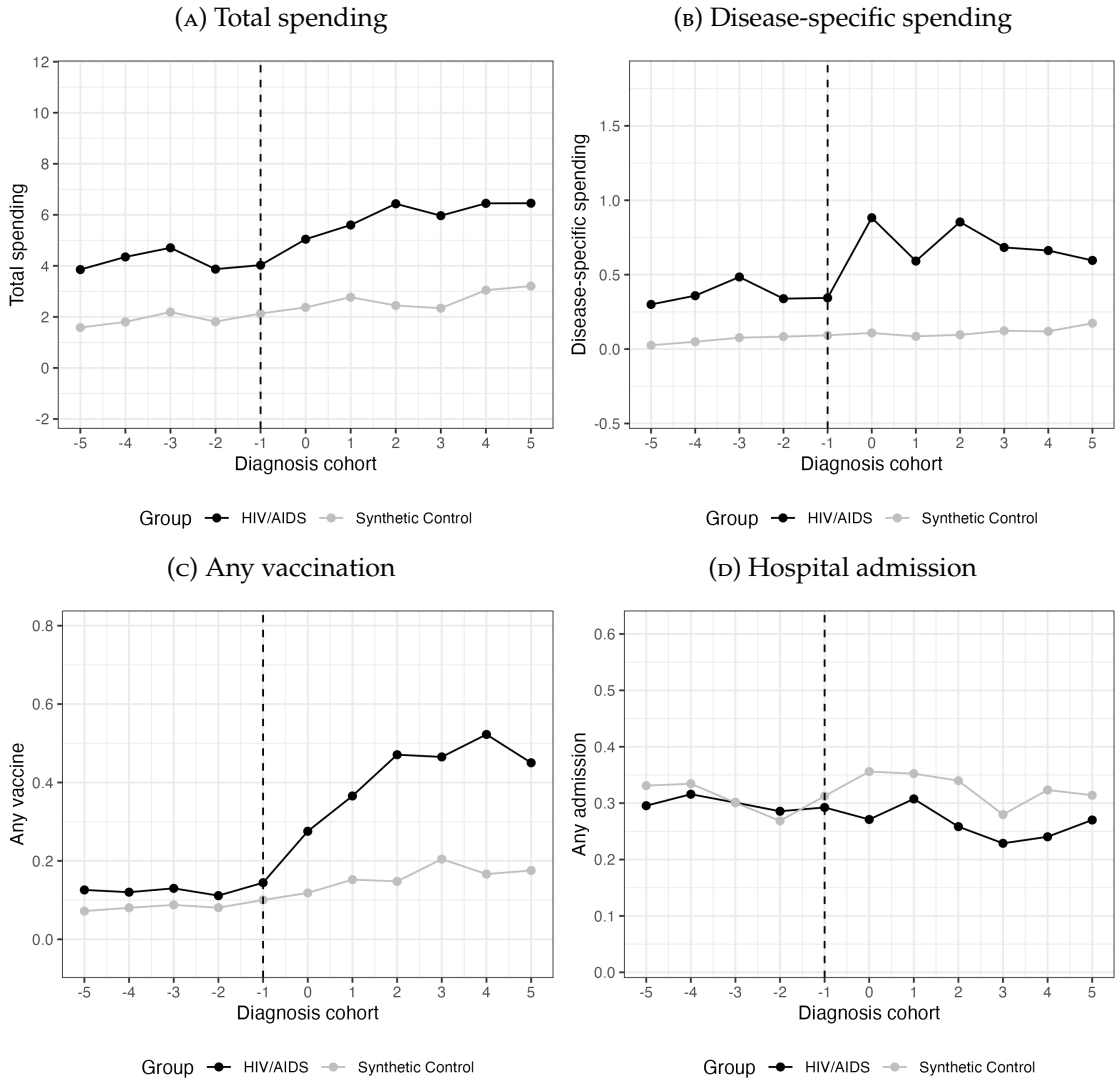
Note: Figure presents the trends of medical spending around diagnosis of cancer, HIV, and control conditions by disease. The dashed lines represent the aggregation window used in the analysis. Cancers affected by HCA are defined as treated and others as controls. See Appendix Table 1 for the ICD codes used.

Appendix Figure 3: Raw Trends for Cancer Patients



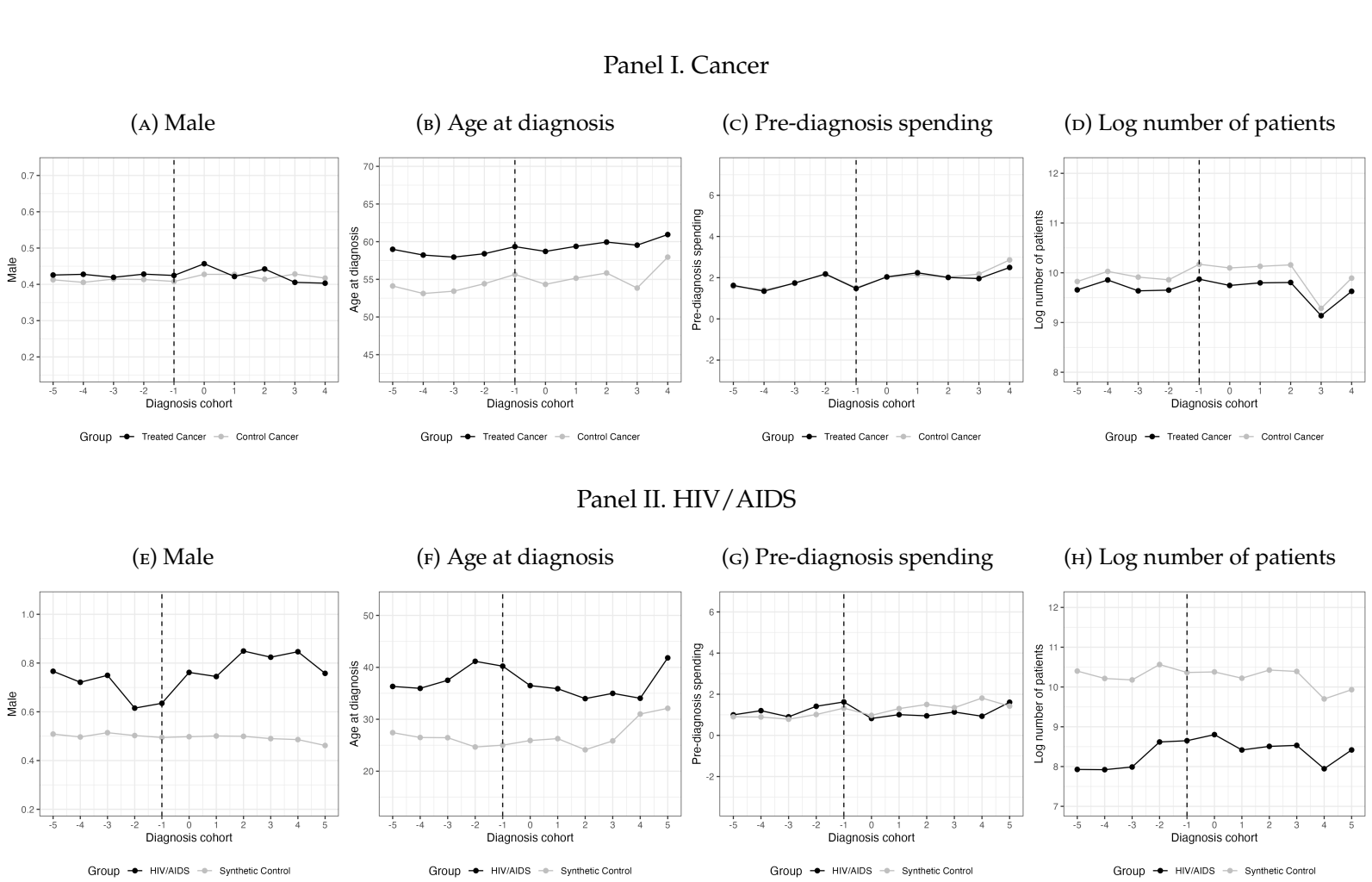
Note: Figure presents the raw mean of each outcome by diagnosis cohort for treated cancer patients (black) and control cancer patients (grey). Outcomes are aggregated over the three years from diagnosis, the same window as the cohort DiD (Equation 2). The dashed line marks the first cohort diagnosed after the HCA policy. Cancer treatment includes chemotherapy, radiotherapy, and cancer-related surgeries (see Appendix Table 2 for details).. Utilization outcomes are measured in the year of the diagnosis and the two years after, where cancer-induced utilization are most concentrated.

Appendix Figure 4: Raw Trends by HIV Patients and Synthetic Control



Note: Figure presents the raw mean of each outcome by diagnosis cohort for HIV patients (black) and the synthetic control (grey). The synthetic control is the mean over control chronic conditions weighted by the ICD-3 synthetic control weights used in the cohort event study (Equation 2). Outcomes are aggregated over the three years from diagnosis. The dashed line marks the first cohort diagnosed after the HCA policy. Antiretroviral therapy (ART) is defined as claims with an ATC code of J05AF/G/E/J/R.

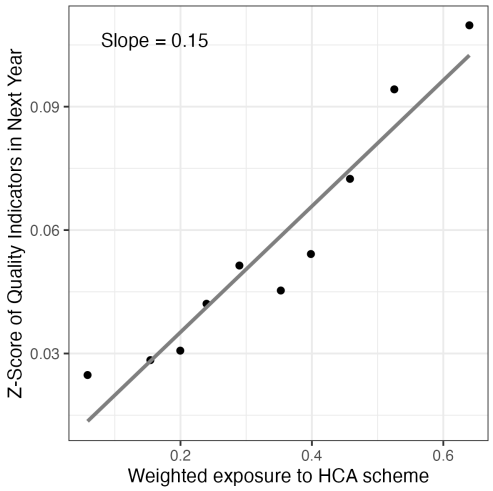
Appendix Figure 5: Placebo Tests for Cancer and HIV/AIDS HCA



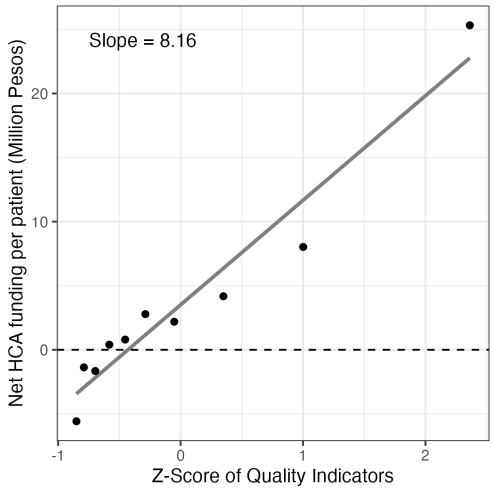
Note: Figure reports placebo outcomes for the cancer (Panel A) and HIV/AIDS (Panel B) event study designs. Pre-diagnosis spending is defined as total spending in the year before diagnosis. Dashed vertical line indicates the start of the perfomance pay policy.

Appendix Figure 6: Predicting Payments and Likelihood of Meeting Quality Thresholds

(a) Intertemporal Correlation of Scores

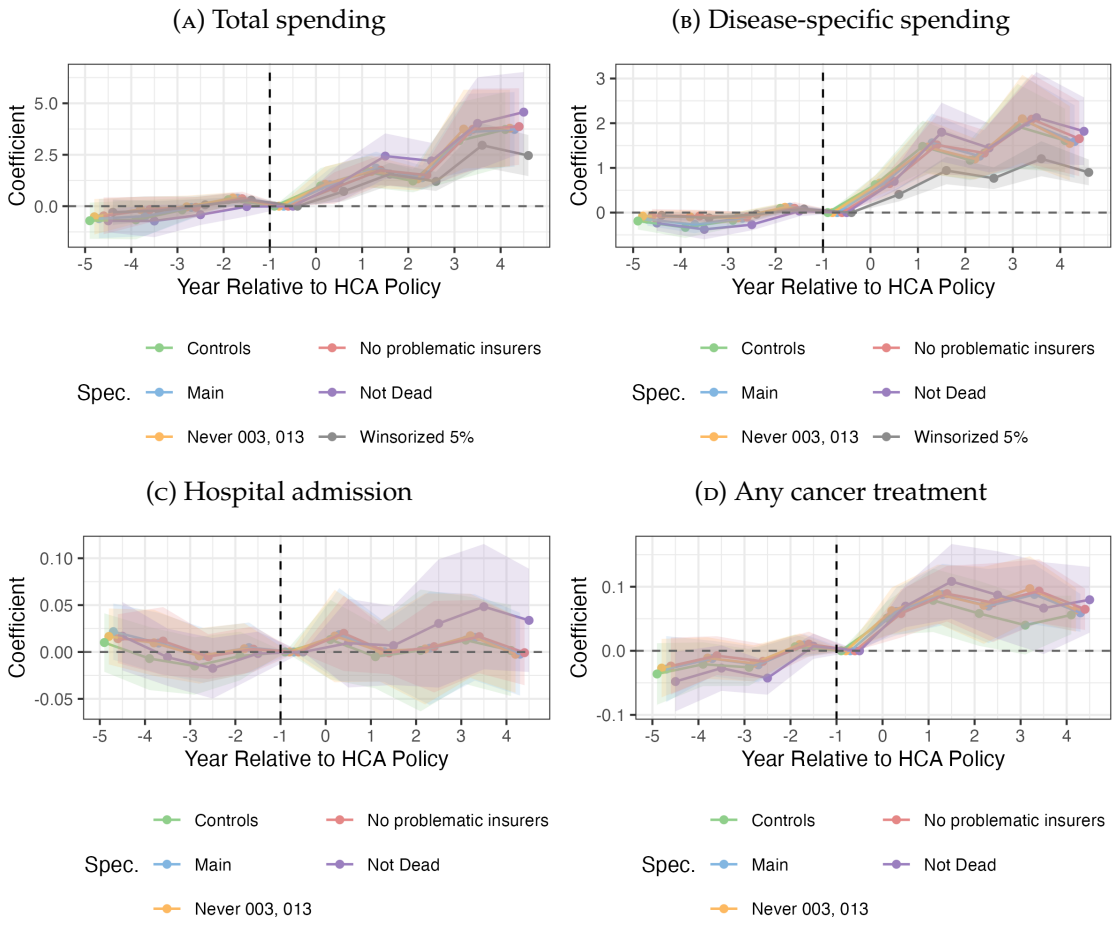


(b) HCA Payments and Quality Scores



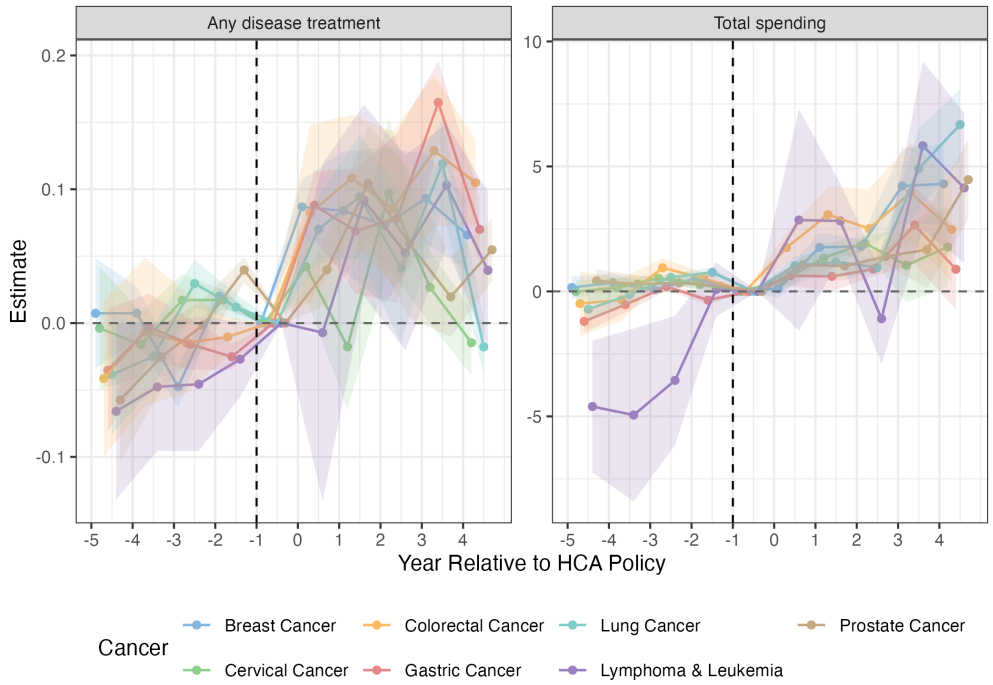
Note: Panel (A) presents the correlation between the likelihood of receiving performance payments (as defined in Equation 1) and the standardized value of the quality indicators in the next year. Panel (B) presents the correlation between standardized quality indicator and net HCA funding per patient in the year. An observation is an insurer-year-cancer. The dots report the conditional means in each bin, the line is a linear fit.

Appendix Figure 7: Robustness Checks on Cancer HCA Impact



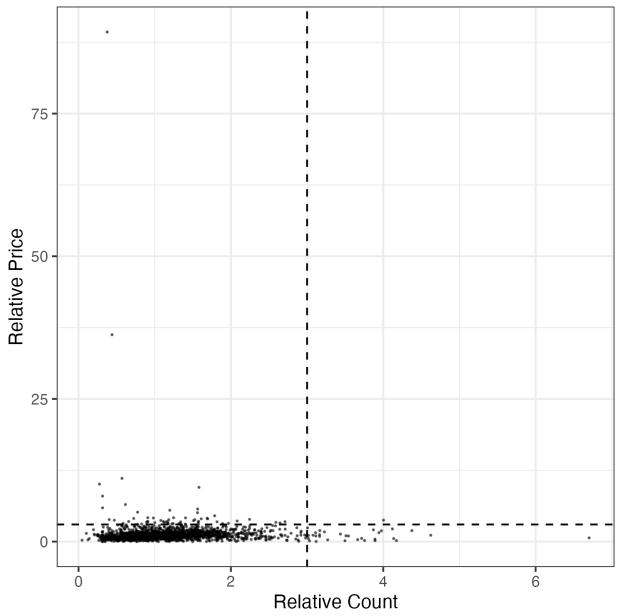
Note: Figure reports the robustness result from DiD design for cancer patients. 'Controls' controls for insurer-year and diagnosis-age-gender fixed effect. 'Never 003, 013' removes patient who ever enrolled with EPS003, EPS013, the two insurers terminated in 2016. 'Not Dead' removes patient who dies in the sample period. 'No problematic insurers' removes patient enrolled with EPS003, EPS013, EPS008, three insurers with a high chance of containing outliers. See note for Figure 4 for details.

Appendix Figure 8: Robustness Checks by Type of Cancer



Note: Figure reports the heterogeintiy result from DiD design for cancer patients by cancer type. We estimate the baseline DiD with the treatment indicator interacted with the type of cancer. See note for Figure 4 for details.

Appendix Figure 9: Outlier Cohorts in the Cancer Sample



Note: Figure reports the relative patient count and cohort cells by Insurer–diagnosis–cohort cells compared to the average of all other years for that insurer–diagnosis combination. The dashed line mark the ratio threshold where observations exceeding the 3 times the relative mean are excluded.

Appendix Table 4: Medicare Advantage Star Quality Rating Measure

	Measure
	Breast Cancer Screening
	Colorectal Cancer Screening
	Annual Flu Vaccine
	Improving or Maintaining Physical Health
	Improving or Maintaining Mental Health
	Monitoring Physical Activity
	Special Needs Plan (SNP) Care Management
	Care for Older Adults – Medication Review
	Care for Older Adults – Pain Assessment
	Osteoporosis Management in Women who had a Fracture
	Diabetes Care – Eye Exam
	Diabetes Care – Blood Sugar Controlled
	Kidney Health Evaluation for Patients with Diabetes
	Controlling Blood Pressure
	Reducing the Risk of Falling
	Improving Bladder Control
	Medication Reconciliation Post-Discharge
	Plan All-Cause Readmissions
	Statin Therapy for Patients with Cardiovascular Disease
	Transitions of Care
	Follow-up after ED Visit for People with Multiple High-Risk Chronic Conditions
	Getting Needed Care
	Getting Appointments and Care Quickly
	Customer Service
	Rating of Health Care Quality
	Rating of Health Plan
	Care Coordination
	Complaints about the Plan
	Members Choosing to Leave the Plan
	Health Plan Quality Improvement
	Plan Makes Timely Decisions about Appeals
	Reviewing Appeals Decisions
	Call Center – Foreign Language Interpreter and TTY Availability

Note: Table reports the quality measures for Star Ratings Part C in Medicare, accessed from https://www.cms.gov/files/document/2027-star-ratings-measures.pdf.